Project1

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1. Introduction

The term 'LEGO set' is a familiar one, associated with fun across generations allowing for kids and adults to build and create alongside one another. Aside from being a household name, LEGO is a global toy brand with a wide variety of products that differ in size, complexity, and price. Understanding how set characteristics are associated with original retail price is relevant to consumers, collectors, and manufacturers interested in LEGO pricing.

The Brickset dataset offers a rich opportunity to explore how features related to price impact the market. In regards to the price, our analysis will use the original US retail price for a LEGO set which was set by the company unlike the resale price established by the person selling a set or the market price which adjusts for how much it would be worth if sold today. Using the original retail price will allow us to draw more accurate conclusions on what characteristics of a set are most associated to its cost as the price is based on the company selling the set, not what others think it is worth. By analyzing these variables and applying model selection techniques, the goal is to identify the most strongly associated with original retail price and construct a model that prioritizes not only predictive accuracy but also interpretability. This analysis will also allow us to examine larger patterns in LEGO set design and pricing across themes.

2. Research Questions

The Brickset LEGO dataset provides detailed information about LEGO sets released in 2025 including price, piece count, themes, and other characteristics. Since sets vary considerably in size and complexity, this dataset provides an opportunity to investigate how these characteristics are associated with retail price and how effectively they can be used to predict a set's original cost.

Our primary research question is:

1. Which LEGO set characteristics are most strongly associated with original US retail price?

To answer this question, we will examine the relationship between original U.S. retail price and several characteristics of LEGO sets released in 2025, including piece count, weight, and other available product features.

We will begin by conducting exploratory data analysis to understand the distributions of the variables, identify potential relationships with retail price, and assess whether any observations or patterns may affect the regression analysis. Next, we will use multiple linear regression to evaluate the associations between set characteristics and retail price. Model selection techniques, including best subsets selection and stepwise regression, will then be used to compare candidate models and identify which combination of predictors provides the best balance between explanatory power and simplicity. We will evaluate the selected model using measures of model fit, such as adjusted R^2, alongside residual plots to assess whether the regression assumptions are reasonably satisfied. Through this process, the analysis aims to identify the characteristics most closely associated with original retail price and develop an interpretable model that explains price variation among LEGO sets released in 2025. As the data is observational, the findings will describe statistical associations rather than establish causal relationships.

3. Data Description

The dataset used in this analysis is from Brickset, primarily known as an online LEGO database that catalogs official LEGO sets released, along with details about minifigures, parts, colors, set inventories, and release information. The dataset was accessed through Github and can be accessed using this link:

https://github.com/jbryer/brickset/blob/main/data-raw/sets/2025.csv

Each row in the dataset represents a single LEGO set and contains measurements recorded at the time of the set's original retail release. These included the sets' US retail price, physical characteristics, packaging dimensions, recommended age range, customer rating information. Because Brickset is a real-world catalog of LEGO products, the dataset consists of authentic observations rather than simulated data.

The full dataset contains 1,079 LEGO sets, spanning multiple themes and product families. The filename "2025.csv" includes LEGO sets that were released in 2025 or close to with some dates in 2024. As the 2024 dates are within one month of 2025, for the purpose of this analysis those sets will be considered as 2025 sets accounting for about one year of LEGO set releases. Retail price is recorded in US dollars, weight is measured in grams, and packaging dimensions(height, width, depth) are recorded in millimeters. The variable pieces represents the number of building elements included in the set, and minifigs refers to the count of included minifigure characters. The dataset also includes several categorical fields, theme, themeGroup, and category, classifying the sets into product types of families.

```r
variable_table <- data.frame(
  Variable = c(
    "US_retailPrice", "pieces", "minifigs", "weight",
    "height", "width", "depth", "agerange_min",
    "rating", "reviewCount", "theme", "themeGroup", "category"
  ),

  Role = c(
    "Response",
    rep("Quantitative Predictor", 9),
    rep("Categorical Predictor", 3)
  ),

  Data_Type = c(
    "Numeric", "Numeric", "Numeric", "Numeric", "Numeric",
    "Numeric", "Numeric", "Numeric", "Numeric", "Numeric",
    "Categorical", "Categorical", "Categorical"
  ),

  Description = c(
    "Original U.S. retail price of the LEGO set in U.S. dollars",
    "Total number of LEGO pieces",
    "Number of minifigures included in the set",
    "Weight of the LEGO set",
    "Height of the LEGO set",
    "Width of the LEGO set",
    "Depth of the LEGO set",
    "Minimum recommended age for the set",
    "Average user rating for the set",
    "Number of user reviews",
    "Specific LEGO theme associated with the set",
    "Broader grouping of LEGO themes",
    "Category or type of LEGO set"
  )
)
variable_table
```

```
         Variable                   Role   Data_Type
1  US_retailPrice               Response     Numeric
2          pieces Quantitative Predictor     Numeric
3        minifigs Quantitative Predictor     Numeric
4          weight Quantitative Predictor     Numeric
```

```
5          height Quantitative Predictor     Numeric
6           width Quantitative Predictor     Numeric
7           depth Quantitative Predictor     Numeric
8    agerange_min Quantitative Predictor     Numeric
9          rating Quantitative Predictor     Numeric
10    reviewCount Quantitative Predictor     Numeric
11          theme  Categorical Predictor Categorical
12     themeGroup  Categorical Predictor Categorical
13       category  Categorical Predictor Categorical
                                                 Description
1  Original U.S. retail price of the LEGO set in U.S. dollars
2                                 Total number of LEGO pieces
3                  Number of minifigures included in the set
4                                      Weight of the LEGO set
5                                      Height of the LEGO set
6                                       Width of the LEGO set
7                                       Depth of the LEGO set
8                        Minimum recommended age for the set
9                            Average user rating for the set
10                                     Number of user reviews
11               Specific LEGO theme associated with the set
12                           Broader grouping of LEGO themes
13                              Category or type of LEGO set
```

4. Data Cleaning

The original dataset contains 1079 LEGO sets. Several variables weight and packaging dimensions have large amounts of missing data, meaning that requiring weight for all observations reduces the usable sample substantially. Other fields, height, width, depth, and reviewCount, also contain missing values.

If the dataset were to only included data points that included all data and contain no missing values, including categorical descriptors, the dataset would be reduced down to 61 complete observations which is 5.7% of the original dataset. Because such a small subset may not be representative of the full collection of LEGO sets, it is necessary to evaluate the consequences of this reduction.

```r
#load data and read csv
df <- read.csv("2025.csv")

#create a table that counts all the missing data
variables <- c(
```

```
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount",
  "theme",
  "themeGroup",
  "category"
)

missing_table <- data.frame(
  Variable = variables,
  Missing = colSums(is.na(df[variables]))
)

missing_table
```

```
                    Variable Missing
US_retailPrice US_retailPrice     448
pieces               pieces     271
minifigs           minifigs     621
weight               weight     893
height               height     421
width                 width     421
depth                 depth     430
agerange_min   agerange_min     270
rating               rating       0
reviewCount     reviewCount       0
theme                 theme       0
themeGroup       themeGroup       1
category           category       0
```

Variables with high coverage, such as retail price, piece count, and minifigure count will be used to compare retained and excluded sets. This will help determine whether the complete-case subset disproportionately contains sets from particular themes, sizes, and price ranges.

Zero values are also examined to see if it counts towards missingness or true measurements. A minifigure count of zero is a valid characteristic for certain sets, and a review count of zero may indicate that no reviews were submitted rather than missing data. These values were retained.

Because requiring weight excludes majority of the dataset, we adopt a staged analysis:

1. High-Coverage Model:

- Uses variables with minimal missingness(price, pieces, minifigs, dimensions). This model retains hundreds of observations and provides a broader view of pricing relationships.

2. Rich but Small Model:

- Uses the all the predictor variables within the complete case set with no missing values, but has limited coverage.

These models will be compared to evaluate how the sample size affect model selection and inference.

To prepare the dataset for analysis, we performed several data cleaning steps to ensure accuracy and consistency across all observations. This process involved identifying missing values and removing incomplete cases. The cleaned dataset, referred to as *data_complete*.

```
#create a new table that omits all the missing data
variables <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount",
  "theme",
  "themeGroup",
  "category"
)

data_complete <- df[complete.cases(df[variables]), ]
nrow(df)
```

```
[1] 1079
```

```r
nrow(data_complete)
```

```
[1] 61
```

```r
colSums(is.na(data_complete[variables]))
```

```
US_retailPrice         pieces       minifigs         weight         height
             0              0              0              0              0
         width          depth   agerange_min         rating    reviewCount
             0              0              0              0              0
         theme     themeGroup       category
             0              0              0
```

5. Exploratory Data Analysis

We begin by examining the distribution of key variables in the full dataset. This analysis will help determine the characteristics of variables and how they impact the model selection process. Correlation between variables is also an important feature to be analyzed, so the relationship between variables can be determined.

```r
# full dataset summaries
summary(df$US_retailPrice)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
   4.99   19.99   39.99   66.34   79.99  999.99     448
```

```r
summary(df$pieces)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
    1.0    42.0   205.5   479.7   582.5  9023.0     271
```

```r
summary(df$minifigs)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
  1.000   1.000   2.000   3.275   4.000  40.000     621
```

```r
summary(df$weight)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
 0.0070  0.1512  0.3255  0.6092  0.6843  3.9030     893
```

```r
summary(df$height)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
   5.20   14.70   19.10   22.69   26.20   55.50     421
```

```r
summary(df$width)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
   7.58   17.00   26.20   27.15   37.80   55.70     421
```

```r
summary(df$depth)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
  0.200   4.500   6.100   6.552   7.600  38.300     430
```

```r
summary(df$rating)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   0.000   0.000   1.055   3.400   4.800
```

```r
summary(df$reviewCount)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.0000  0.0000  0.1835  0.0000  4.0000
```

```r
# complete case subset summaries from data_complete
summary(data_complete$US_retailPrice)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   9.99   19.99   39.99   58.92   79.99  299.99
```

```r
summary(data_complete$pieces)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   42.0   154.0   261.0   518.7   678.0  3266.0
```

```r
summary(data_complete$minifigs)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00    2.00    3.00    3.23    4.00   10.00
```

```r
summary(data_complete$weight)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.1200  0.3260  0.5210  0.9087  1.2340  3.9030
```

```r
summary(data_complete$height)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  12.20   19.10   26.00   24.01   28.20   48.00
```

```r
summary(data_complete$width)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  14.10   20.50   28.20   32.22   45.60   55.50
```

```r
summary(data_complete$depth)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   4.60    6.10    6.20    7.33    8.80   13.20
```

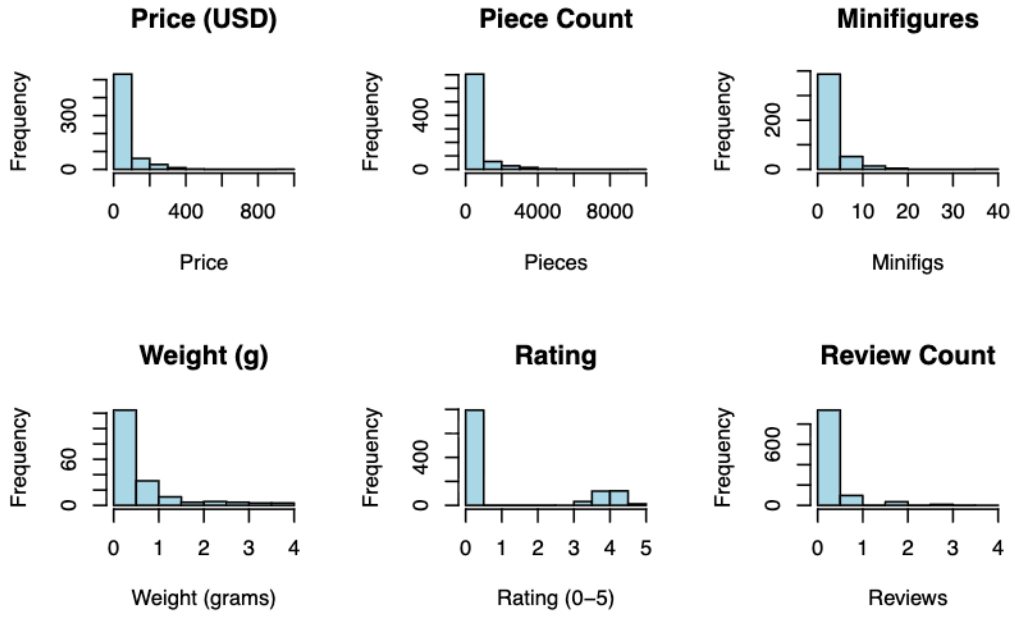
```r
summary(data_complete$rating)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   0.000   0.000   1.584   3.800   4.500
```

```
summary(data_complete$reviewCount)
```

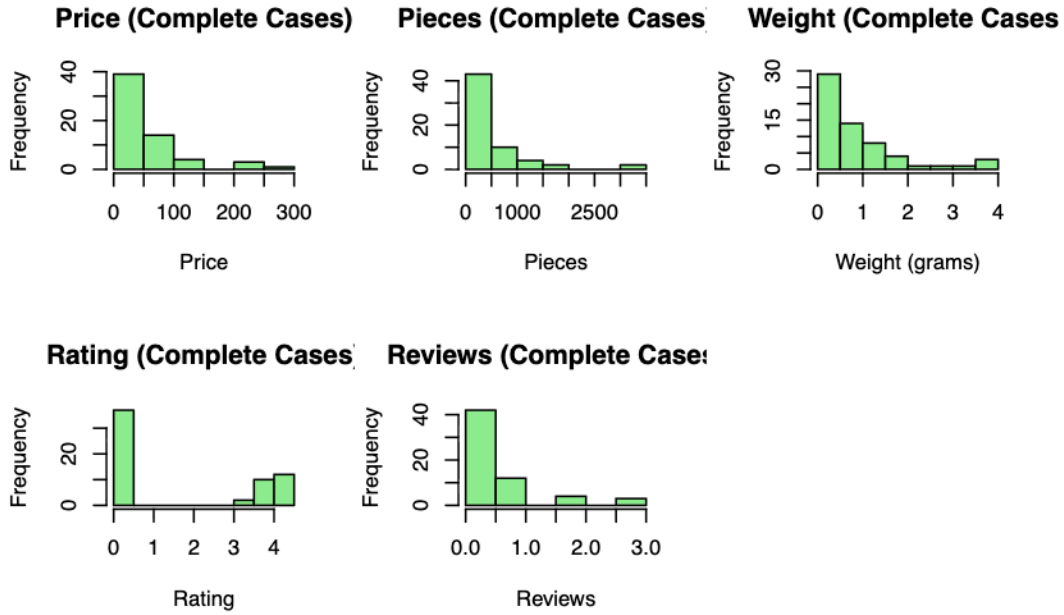
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.0000  0.0000  0.4754  1.0000  3.0000
```

```
#Histograms for full dataset vs the cleaned dataset
par(mfrow = c(2, 3))
hist(df$US_retailPrice, main = "Price (USD)", xlab = "Price", col = "lightblue")
hist(df$pieces, main = "Piece Count", xlab = "Pieces", col = "lightblue")
hist(df$minifigs, main = "Minifigures", xlab = "Minifigs", col = "lightblue")
hist(df$weight, main = "Weight (g)", xlab = "Weight (grams)", col = "lightblue")
hist(df$rating, main = "Rating", xlab = "Rating (0-5)", col = "lightblue")
hist(df$reviewCount, main = "Review Count", xlab = "Reviews", col = "lightblue")
```




```
# Histograms (61-case subset)
par(mfrow = c(2, 3))
hist(data_complete$US_retailPrice, main = "Price (Complete Cases)",
     xlab = "Price", col = "lightgreen")
hist(data_complete$pieces, main = "Pieces (Complete Cases)",
     xlab = "Pieces", col = "lightgreen")
hist(data_complete$weight, main = "Weight (Complete Cases)",
```

```
     xlab = "Weight (grams)", col = "lightgreen")
hist(data_complete$rating, main = "Rating (Complete Cases)",
     xlab = "Rating", col = "lightgreen")
hist(data_complete$reviewCount, main = "Reviews (Complete Cases)",
     xlab = "Reviews", col = "lightgreen")
```



Distribution in Full Dataset

Retail price shows right-skewness where most LEGO sets are clustered at lower prices and a long right tail where LEGO set prices increases and occurs less. The median price is $39.99, but the means rises to $66.34, showing that there is a small number of expensive sets that pull the average price upward. Prices range from $4.99 to $999.99 with 448 sets missing price.

Piece count has similar behavior. The median set contains about 205.5 pieces, but the mean is about 479.7 pieces, showing a long right tail. There are clusters of LEGO sets with a smaller amount of pieces, and trails off as the amount of pieces increase. Piece count ranges from 1 to 9,023, with 271 missing values.

Minifigure counts also vary widely. Although the median is 2 minifigures, the mean is 3.275, and counts range from 1 to 40. 621 sets have missing minifigure data, and many sets include zero minifigures.

Weight has the highest missingness as 893 sets lack weight data, leaving only 186 sets with recorded weights. This extremely high missingness explains why complete-case subset shrinks so dramatically.

Packaging dimensions (height, width, depth) also show substantial missingness (421–430 missing values each). Among observed values, height ranges from 5.2 mm to 55.5 mm, width from 7.58 mm to 55.7 mm, and depth from 0.2 mm to 38.3 mm. These variables are right-skewed and strongly correlated with piece count and weight.

Ratings and review counts are heavily zero-inflated. The median rating is 0, and the median review count is 0, indicating that many sets have no user feedback recorded. Ratings range from 0 to 4.8, and review counts from 0 to 4.

Distribution within the Complete-Case Subset

The complete-case subset differs noticeably from the full dataset. Price remains right-skewed, but the range is narrower: \$9.99 to \$299.99, with a mean of \$58.92. This suggests that the complete cases under-represent very high-priced sets.

Piece counts also show reduced range, 42 to 3,266 pieces, compared to the full dataset's maximum of 9,023. The median piece count is 261, slightly higher than the full dataset median (205.5), indicating that the complete-case subset may over-represent medium-sized sets.

Minifigure counts are more concentrated: median 3, mean 3.23, and maximum 10, compared to 40 in the full dataset. This again suggests that the complete-case subset excludes many large or specialized sets.

Weight is fully observed in this subset by construction, with a median of 0.521 kg and mean of 0.9087 kg, substantially higher than the full dataset's median (0.3255 kg). This indicates that sets with recorded weight tend to be larger and heavier.

Dimensions also show higher averages in the complete-case subset. For example, median height 26 mm, median width 28.2 mm. This reinforces the idea that complete cases disproportionately include larger sets.

Ratings and review counts have higher means in the complete-case subset (mean rating 1.584, mean review count 0.475), suggesting that sets with complete metadata may be more popular or more frequently reviewed.

Key Takeaways

- Strong right-skewness in price, pieces, weight, and dimensions
- Zeros in ratings and review counts are to be interpreted as "no reviews" rather than missing data
- Weight and dimensions have a high amount of missing data, which is why the complete-case subset only leaves 61 LEGO sets to be observed
- Based on the summaries, the complete-case subset is not representative of the data because it contains heavier, larger sets, it excludes many high priced and high piece count LEGO sets, and it over represents sets with more complete data and reviews.

Correlation Analysis

```
# quantitative variables

quantitative_vars <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount"
)

# Select quantitative variables
correlation_vars <- data_complete[quantitative_vars]

# Calculate the correlation matrix
correlation_matrix <- cor(
  correlation_vars,
  use = "complete.obs"
)

correlation_matrix
```

```
              US_retailPrice    pieces  minifigs    weight    height     width
US_retailPrice      1.0000000 0.9106009 0.7786630 0.9671660 0.8191912 0.7984531
pieces              0.9106009 1.0000000 0.7555396 0.8658698 0.6260305 0.6528074
minifigs            0.7786630 0.7555396 1.0000000 0.7486760 0.6631737 0.6829421
weight              0.9671660 0.8658698 0.7486760 1.0000000 0.8464519 0.8272257
height              0.8191912 0.6260305 0.6631737 0.8464519 1.0000000 0.8323138
width               0.7984531 0.6528074 0.6829421 0.8272257 0.8323138 1.0000000
depth               0.7667257 0.6624670 0.5304681 0.8274525 0.7266921 0.7192446
agerange_min        0.5728106 0.7147580 0.3859962 0.4524908 0.3079079 0.3201698
rating              0.2523957 0.3476234 0.2676433 0.2636961 0.1064871 0.1294318
reviewCount         0.2599065 0.3492340 0.2701237 0.2828723 0.1505186 0.1648387
                    depth agerange_min    rating reviewCount
US_retailPrice 0.7667257    0.5728106 0.2523957   0.2599065
pieces         0.6624670    0.7147580 0.3476234   0.3492340
```

```
minifigs       0.5304681    0.3859962 0.2676433   0.2701237
weight         0.8274525    0.4524908 0.2636961   0.2828723
height         0.7266921    0.3079079 0.1064871   0.1505186
width          0.7192446    0.3201698 0.1294318   0.1648387
depth          1.0000000    0.4066463 0.1718117   0.2724942
agerange_min   0.4066463    1.0000000 0.3509209   0.3342968
rating         0.1718117    0.3509209 1.0000000   0.5944375
reviewCount    0.2724942    0.3342968 0.5944375   1.0000000
```

The correlation matrix shows strong positive relationships between LEGO set retail price and several measures of set size and complexity. The strongest correlation is between US retail price and weight ($r = 0.967$), followed by the number of pieces ($r = 0.911$). This suggests that heavier sets and sets containing more pieces tend to have higher retail prices. Price also has strong positive correlations with height ($r = 0.819$), width ($r = 0.798$), depth ($r = 0.767$), and the number of minifigures ($r = 0.779$). These results suggest that the physical size and contents of a LEGO set are closely associated with its retail price. The minimum recommended age also has a moderate positive correlation with price ($r = 0.573$), indicating that sets designed for older builders may tend to be more expensive. In contrast, user ratings ($r = 0.252$) and review counts ($r = 0.260$) have relatively weak correlations with price, suggesting that customer ratings and the number of reviews are less strongly associated with retail price than the physical characteristics of a set.

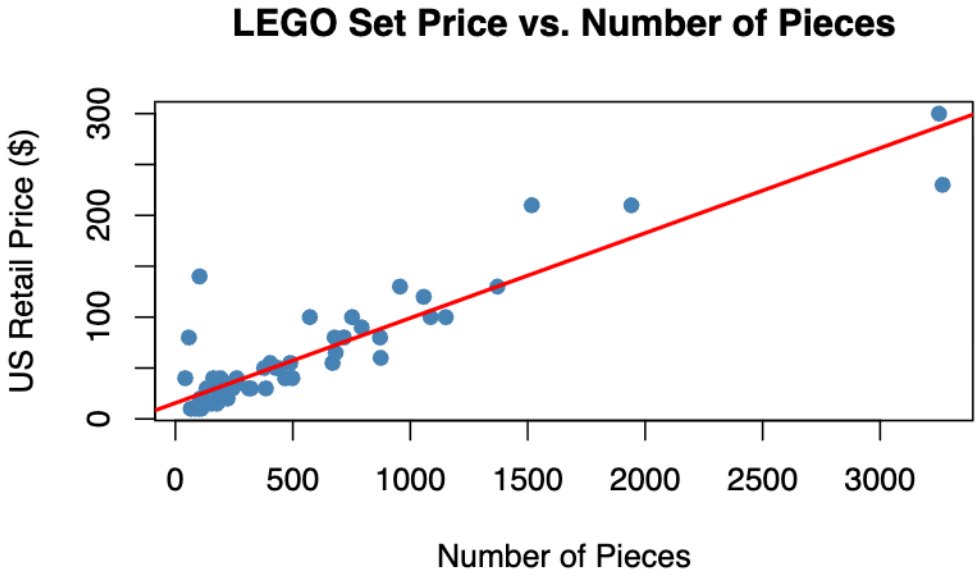
The matrix also reveals strong correlations among several predictor variables. For example, piece count and weight have a correlation of 0.866, while weight is also strongly correlated with height ($r = 0.846$), width ($r = 0.827$), and depth ($r = 0.827$). These relationships suggest that these variables could create multicollinearity in a multiple linear regression model. This means that including all these predictors simultaneously may make it harder to distinguish their individual contributions to price. Additionally, ratings and review counts have a moderate positive correlation ($r = 0.594$), suggesting that sets with more reviews may also tend to have higher ratings. Overall, the correlation matrix suggests that weight and piece count are promising predictors of retail price, although further regression analysis and model selection will be necessary to determine which variables provide the most useful independent explanatory power.

```r
#Scatter plots of key relationships

plot(
  US_retailPrice ~ pieces,
  data = data_complete,
  main = "LEGO Set Price vs. Number of Pieces",
  xlab = "Number of Pieces",
  ylab = "US Retail Price ($)",
  pch = 19,
```

```
  col = "steelblue"
)

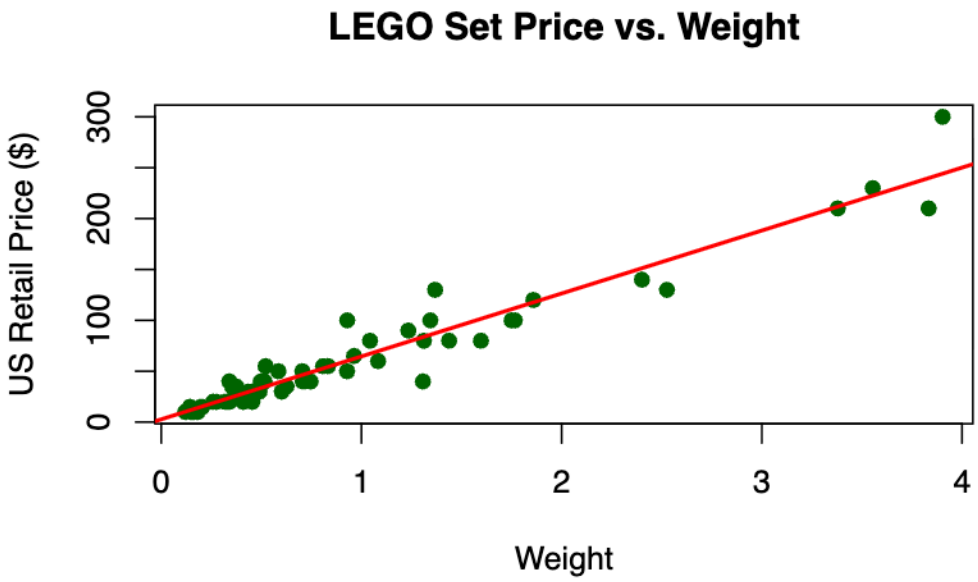
# Add a fitted linear regression line
abline(
  lm(US_retailPrice ~ pieces, data = data_complete),
  col = "red",
  lwd = 2
)
```




```
plot(
  US_retailPrice ~ weight,
  data = data_complete,
  main = "LEGO Set Price vs. Weight",
  xlab = "Weight",
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "darkgreen"
)

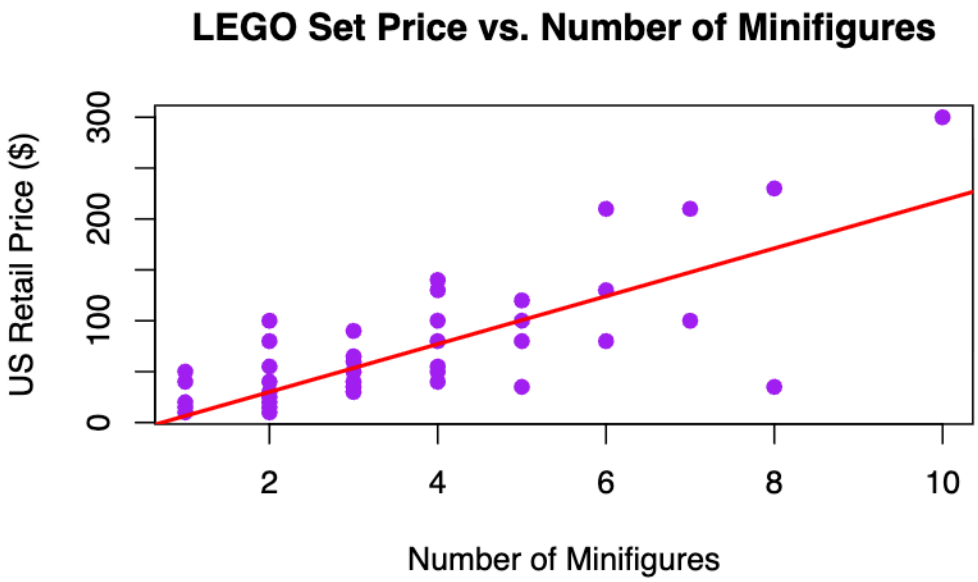
abline(
  lm(US_retailPrice ~ weight, data = data_complete),
```

```
  col = "red",
  lwd = 2
)
```




```
plot(
  US_retailPrice ~ minifigs,
  data = data_complete,
  main = "LEGO Set Price vs. Number of Minifigures",
  xlab = "Number of Minifigures",
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "purple"
)

abline(lm(US_retailPrice ~ minifigs, data = data_complete),col = "red",lwd = 2)
```



The scatterplots reveal positive relationships between LEGO set retail price and the number of pieces and weight. The strongest visual relationship appears in the plot of retail price versus weight, where price generally increases as the weight of a set increases. This is consistent with the correlation coefficient of 0.967, indicating a very strong positive linear association. The plot of retail price versus number of pieces also shows a clear upward trend, consistent with its correlation of 0.911. Sets with more pieces generally have higher retail prices, although there is some variation among sets with similar piece counts. Several observations lie farther from the regression line, suggesting that some sets are considerably more expensive or less expensive than their piece counts alone might predict. These observations may reflect differences in set design, theme, or other characteristics that influence pricing. The plots support the correlation analysis by indicating that physical characteristics, particularly weight and piece count, are strongly associated with LEGO set retail price. However, the presence of outlying observations and variation around the regression lines suggests that additional variables may help explain differences in price.

The relationship between retail price and the number of minifigures is also positive, but the pattern is less consistent than those observed for weight and piece count. Although sets with more minifigures tend to have higher prices, there is considerable variation in price among sets containing the same number of minifigures. For example, sets containing six minifigures have substantially different retail prices, suggesting that minifigure count alone does not fully explain a set's price. Overall, the scatterplots support the correlation analysis by indicating that physical characteristics, particularly weight and piece count, are strongly associated with LEGO set retail price. However, the presence of outlying observations and variation around the regression lines suggests that additional variables may help explain differences in price.

6. Best Subsets Regression

Best Subsets Regression is a model selecting method that takes the response variable, in this case US_retailPrice, and the predictor variables within the dataset and yields the possible models and their statistics. This method evaluates every possible combination of predictors and identifies the best model for each model size using adjusted R-squared, BIC and Cp. Adjusted R-squared accounts for model size to tell us how well the predictors explain the variation in the response variable, adding a weak predictor can cause it to drop. BIC penalizes adding parameters which tells us how simple and interpretable a model is; the lower the BIC the better the model. Cp tells how well a model balances fit and complexity; the lower the Cp the better the model.

Stage 1 - Large sample

Because several predictors in the full dataset have substantial missingness, especially weight, rating, and reviewCount, we pay attention to variables with relatively high coverage: pieces, minifigs, height, width, depth, and agerange_min. These variables allow us to retain a much larger portion of the dataset while still capturing key physical characteristics of LEGO sets.

```
# load the package
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tidyr)

# counts the amount of observations(rows) in Stage 1
df_stage1 <- df %>% drop_na(US_retailPrice, pieces, minifigs,
                            height, width, depth, agerange_min)

nrow(df_stage1)
```

```
[1] 289
```

After removing rows with missing values in these fields, the Stage 1 dataset contains 289 observations, more than the 61 complete cases available when weight and rating are required. This larger sample provides a more representative view of the overall LEGO product catalog and serves as the foundation for the high-coverage model selection.

```r
 # load package for best subsets regression
library(leaps)

# run best subsets
best_subsets <- regsubsets(US_retailPrice ~ pieces + minifigs +
                             height + depth + width + agerange_min,
                           data = df_stage1)


# display summary of the best regression models
best_summary_model <- summary(best_subsets)
best_summary_model
```

```
Subset selection object
Call: regsubsets.formula(US_retailPrice ~ pieces + minifigs + height +
    depth + width + agerange_min, data = df_stage1)
6 Variables  (and intercept)
             Forced in Forced out
pieces           FALSE      FALSE
minifigs         FALSE      FALSE
height           FALSE      FALSE
depth            FALSE      FALSE
width            FALSE      FALSE
agerange_min     FALSE      FALSE
1 subsets of each size up to 6
Selection Algorithm: exhaustive
         pieces minifigs height depth width agerange_min
1  ( 1 ) "*"    " "      " "    " "   " "   " "
2  ( 1 ) "*"    " "      " "    "*"   " "   " "
3  ( 1 ) "*"    "*"      " "    "*"   " "   " "
4  ( 1 ) "*"    "*"      "*"    "*"   " "   " "
5  ( 1 ) "*"    "*"      "*"    "*"   " "   "*"
6  ( 1 ) "*"    "*"      "*"    "*"   "*"   "*"
```

```
best_summary_model$adjr2
```

```
[1] 0.9263590 0.9414897 0.9521875 0.9568381 0.9566949 0.9565445
```

```
best_summary_model$cp
```

```
[1] 201.358340 102.081950  32.574978   3.080734   5.019948   7.000000
```

```
best_summary_model$bic
```

```
[1] -743.5443 -805.3563 -859.0556 -883.9784 -878.3743 -872.7283
```

```
#which tells which predictors are included in each model
best_summary_model$which
```

```
  (Intercept) pieces minifigs height depth width agerange_min
1        TRUE   TRUE    FALSE  FALSE FALSE FALSE        FALSE
2        TRUE   TRUE    FALSE  FALSE  TRUE FALSE        FALSE
3        TRUE   TRUE     TRUE  FALSE  TRUE FALSE        FALSE
4        TRUE   TRUE     TRUE   TRUE  TRUE FALSE        FALSE
5        TRUE   TRUE     TRUE   TRUE  TRUE FALSE         TRUE
6        TRUE   TRUE     TRUE   TRUE  TRUE  TRUE         TRUE
```

Applying the best-subsets regression to this 289 subset using price as the response variable, the four-predictor model with pieces, minifigs, height, and depth is the best model. This model achieves the highest adjusted R-squared (0.9568), indicating the strongest explanatory power among the other models. It also has the lowest Cp(3.08), which is close to the ideal value of p(predictors) + 1 = 5, and the lowest BIC(-883.98), showing the best balance between fit and model complexity. Although, the fifth and sixth predictor models have adjusted R-squared values that are close to the four-predictor model;however, their Cp and BIC values are worse, indicating more complexity by adding more predictors with little improvement in the models explanatory power. Therefore, the four-predictor model, model 4, is the preferred Stage 1 model.

Stage 2 - Complete Case

Stage 2 uses a wider range of the predictor variables but only uses LEGO sets where there are no missing values. This subset is considered rich but small as it only contains 61 observations. In this subset we are including weight because it has strong explanatory value and rating and reviewCount may matter, but because these variables have huge missingness, they must be evaluated separately from Stage 1.

```
df_stage2 <- df %>% drop_na(US_retailPrice, pieces, minifigs, height, width,
                            depth, agerange_min, weight, rating, reviewCount)

nrow(df_stage2)
```

```
[1] 61
```

```
 # load package for best subsets regression
library(leaps)

# run best subsets
best_subsets <- regsubsets(US_retailPrice ~ pieces + minifigs +
                             height + depth + width + agerange_min +
                             weight + rating + reviewCount,
                           data = df_stage2, nvmax = 9)


# display summary of the best regression models
best_summary_model <- summary(best_subsets)
best_summary_model
```

```
Subset selection object
Call: regsubsets.formula(US_retailPrice ~ pieces + minifigs + height +
    depth + width + agerange_min + weight + rating + reviewCount,
    data = df_stage2, nvmax = 9)
9 Variables  (and intercept)
             Forced in Forced out
pieces           FALSE      FALSE
minifigs         FALSE      FALSE
height           FALSE      FALSE
depth            FALSE      FALSE
width            FALSE      FALSE
agerange_min     FALSE      FALSE
```

```
weight           FALSE      FALSE
rating           FALSE      FALSE
reviewCount      FALSE      FALSE
1 subsets of each size up to 9
Selection Algorithm: exhaustive
         pieces minifigs height depth width agerange_min weight rating
1  ( 1 ) " "    " "      " "    " "   " "   " "          "*"    " "
2  ( 1 ) " "    " "      " "    " "   " "   "*"          "*"    " "
3  ( 1 ) " "    "*"      " "    " "   " "   "*"          "*"    " "
4  ( 1 ) " "    "*"      " "    " "   " "   "*"          "*"    " "
5  ( 1 ) " "    "*"      " "    "*"   " "   "*"          "*"    "*"
6  ( 1 ) " "    "*"      " "    "*"   " "   "*"          "*"    "*"
7  ( 1 ) "*"    "*"      " "    "*"   " "   "*"          "*"    "*"
8  ( 1 ) "*"    "*"      "*"    "*"   " "   "*"          "*"    "*"
9  ( 1 ) "*"    "*"      "*"    "*"   "*"   "*"          "*"    "*"
         reviewCount
1  ( 1 ) " "
2  ( 1 ) " "
3  ( 1 ) " "
4  ( 1 ) "*"
5  ( 1 ) " "
6  ( 1 ) "*"
7  ( 1 ) "*"
8  ( 1 ) "*"
9  ( 1 ) "*"
```

```
best_summary_model$adjr2
```

```
[1] 0.9343154 0.9569526 0.9614188 0.9643101 0.9667275 0.9669445 0.9666173
[8] 0.9666360 0.9659913
```

```
best_summary_model$cp
```

```
[1] 56.953050 18.415108 11.663737  7.768334  4.809390  5.486479  7.024448
[8]  8.014201 10.000000
```

```
best_summary_model$bic
```

```
[1] -158.8998 -181.6080 -185.2398 -186.9604 -188.2270 -185.6345 -182.0630
[8] -179.1483 -175.0544
```

```r
#which tells which predictors are included in each model
best_summary_model$which
```

```
  (Intercept) pieces minifigs height depth width agerange_min weight rating
1        TRUE  FALSE    FALSE  FALSE FALSE FALSE        FALSE   TRUE  FALSE
2        TRUE  FALSE    FALSE  FALSE FALSE FALSE         TRUE   TRUE  FALSE
3        TRUE  FALSE     TRUE  FALSE FALSE FALSE         TRUE   TRUE  FALSE
4        TRUE  FALSE     TRUE  FALSE FALSE FALSE         TRUE   TRUE  FALSE
5        TRUE  FALSE     TRUE  FALSE  TRUE FALSE         TRUE   TRUE   TRUE
6        TRUE  FALSE     TRUE  FALSE  TRUE FALSE         TRUE   TRUE   TRUE
7        TRUE   TRUE     TRUE  FALSE  TRUE FALSE         TRUE   TRUE   TRUE
8        TRUE   TRUE     TRUE   TRUE  TRUE FALSE         TRUE   TRUE   TRUE
9        TRUE   TRUE     TRUE   TRUE  TRUE  TRUE         TRUE   TRUE   TRUE
  reviewCount
1       FALSE
2       FALSE
3       FALSE
4        TRUE
5       FALSE
6        TRUE
7        TRUE
8        TRUE
9        TRUE
```

Based on the best subsets regression for stage 2, Model 6 has the highest adjusted R-squared value (0.9669), indicating that it has the strongest explanatory prowess with Model 5, 7, and 8 all being around ~ 0.966. The model that balances fit and complexity the best is Model 5 with a Cp of 4.81, and it also has the best BIC of -188.23. Although Model 6 has the best adjusted R-squared, Model 5 only differs from Model 6 adjusted R-squared by 0.000217. The difference between the models is extremely small, and adding another predictor adds more complexity. Therefore, Model 5 keeps simplicity without losing key explanatory power.

```r
# Define Model 4: Stage 1 best model
model4 <- lm(
  US_retailPrice ~ pieces + minifigs + height + depth,
  data = df_stage1
)

# Define Model 5: Stage 2 best model
model5 <- lm(
  US_retailPrice ~ minifigs + depth + agerange_min + weight + rating,
```

```
  data = df_stage2
)

# Display model summaries
summary(model4)
```

```
Call:
lm(formula = US_retailPrice ~ pieces + minifigs + height + depth,
    data = df_stage1)

Residuals:
    Min      1Q  Median      3Q     Max
-71.426  -9.972  -0.393   8.761  73.052

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -48.675643   5.174107  -9.408  < 2e-16 ***
pieces        0.060727   0.002872  21.144  < 2e-16 ***
minifigs      4.301790   0.498088   8.637 4.25e-16 ***
height        0.939902   0.166915   5.631 4.30e-08 ***
depth         5.887810   0.739335   7.964 4.05e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.13 on 284 degrees of freedom
Multiple R-squared:  0.9574,    Adjusted R-squared:  0.9568
F-statistic:  1597 on 4 and 284 DF,  p-value: < 2.2e-16
```

```
summary(model5)
```

```
Call:
lm(formula = US_retailPrice ~ minifigs + depth + agerange_min +
    weight + rating, data = df_stage2)

Residuals:
    Min      1Q  Median      3Q     Max
-26.482  -5.907  -0.974   5.150  36.940

Coefficients:
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.3865     7.4869  -0.052   0.9590
minifigs       2.7778     1.1134   2.495   0.0156 *
depth         -2.8752     1.1516  -2.497   0.0156 *
agerange_min   2.9673     0.4384   6.768 9.04e-09 ***
weight        58.8252     3.5662  16.495  < 2e-16 ***
rating        -1.8362     0.7603  -2.415   0.0191 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.81 on 55 degrees of freedom
Multiple R-squared:  0.9695,    Adjusted R-squared:  0.9667
F-statistic: 349.7 on 5 and 55 DF,  p-value: < 2.2e-16
```

Backwards Selection

To ensure both models 4 and 5 are the best fitted models for our data, we will perform a backwards selection to confirm.

```
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```
# Stage 1: use the same six candidate predictors
full_stage1 <- lm(
  US_retailPrice ~ pieces + minifigs + height +
    depth + width + agerange_min,
  data = df_stage1
)

backward_stage1 <- step(
  full_stage1,
  direction = "backward",
  trace = 1
)
```

```
Start:  AIC=1714.68
US_retailPrice ~ pieces + minifigs + height + depth + width +
    agerange_min

               Df Sum of Sq    RSS    AIC
- width         1         7 103901 1712.7
- agerange_min  1        21 103914 1712.7
<none>                      103893 1714.7
- height        1      9731 113624 1738.5
- depth         1     20429 124322 1764.6
- minifigs      1     24422 128316 1773.7
- pieces        1     98693 202587 1905.7

Step:  AIC=1712.7
US_retailPrice ~ pieces + minifigs + height + depth + agerange_min

               Df Sum of Sq    RSS    AIC
- agerange_min  1        22 103923 1710.8
<none>                      103901 1712.7
- height        1     11534 115434 1741.1
- depth         1     20423 124324 1762.6
- minifigs      1     24563 128464 1772.0
- pieces        1    100836 204736 1906.7

Step:  AIC=1710.76
US_retailPrice ~ pieces + minifigs + height + depth

           Df Sum of Sq    RSS    AIC
<none>                  103923 1710.8
- height    1     11603 115526 1739.3
- depth     1     23207 127130 1767.0
- minifigs  1     27295 131218 1776.2
- pieces    1    163597 267520 1982.0
```

```
summary(backward_stage1)
```

```
Call:
lm(formula = US_retailPrice ~ pieces + minifigs + height + depth,
    data = df_stage1)

Residuals:
```

```
    Min      1Q  Median      3Q     Max
-71.426  -9.972  -0.393   8.761  73.052

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -48.675643   5.174107  -9.408  < 2e-16 ***
pieces        0.060727   0.002872  21.144  < 2e-16 ***
minifigs      4.301790   0.498088   8.637 4.25e-16 ***
height        0.939902   0.166915   5.631 4.30e-08 ***
depth         5.887810   0.739335   7.964 4.05e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.13 on 284 degrees of freedom
Multiple R-squared:  0.9574,    Adjusted R-squared:  0.9568
F-statistic:  1597 on 4 and 284 DF,  p-value: < 2.2e-16
```

```r
formula(backward_stage1)
```

```
US_retailPrice ~ pieces + minifigs + height + depth
```

```r
# Stage 2: use the same nine candidate predictors
full_stage2 <- lm(
  US_retailPrice ~ pieces + minifigs + height +
    depth + width + agerange_min + weight +
    rating + reviewCount,
  data = df_stage2
)

backward_stage2 <- step(
  full_stage2,
  direction = "backward",
  trace = 1
)
```

```
Start:  AIC=300.82
US_retailPrice ~ pieces + minifigs + height + depth + width +
    agerange_min + weight + rating + reviewCount

               Df Sum of Sq     RSS    AIC
- width         1       1.7  6092.6 298.84
```

```
- height        1     117.9  6208.9 299.99
- pieces        1     135.7  6226.7 300.17
- reviewCount   1     157.0  6247.9 300.38
- rating        1     175.8  6266.7 300.56
<none>                       6091.0 300.82
- minifigs      1     352.1  6443.0 302.25
- depth         1     448.5  6539.5 303.16
- agerange_min  1    1476.0  7567.0 312.06
- weight        1    6786.3 12877.3 344.49

Step:  AIC=298.84
US_retailPrice ~ pieces + minifigs + height + depth + agerange_min +
    weight + rating + reviewCount

               Df Sum of Sq     RSS    AIC
- height        1     120.7  6213.3 298.04
- pieces        1     140.6  6233.3 298.23
- reviewCount   1     155.9  6248.5 298.38
- rating        1     174.5  6267.2 298.56
<none>                       6092.6 298.84
- minifigs      1     358.7  6451.4 300.33
- depth         1     458.5  6551.1 301.27
- agerange_min  1    1474.8  7567.5 310.07
- weight        1    7065.5 13158.2 343.81

Step:  AIC=298.04
US_retailPrice ~ pieces + minifigs + depth + agerange_min + weight +
    rating + reviewCount

               Df Sum of Sq     RSS    AIC
- pieces        1      55.2  6268.5 296.58
- reviewCount   1     168.8  6382.1 297.67
<none>                       6213.3 298.04
- rating        1     228.5  6441.8 298.24
- depth         1     470.6  6683.9 300.49
- minifigs      1     566.6  6779.9 301.36
- agerange_min  1    1939.1  8152.4 312.61
- weight        1   14951.5 21164.8 370.80

Step:  AIC=296.58
US_retailPrice ~ minifigs + depth + agerange_min + weight + rating +
    reviewCount
```

```
               Df Sum of Sq   RSS    AIC
- reviewCount   1     158.0  6426 296.10
<none>                       6268 296.58
- rating        1     231.3  6500 296.79
- depth         1     616.7  6885 300.30
- minifigs      1     769.4  7038 301.64
- agerange_min  1    5480.6 11749 332.90
- weight        1   31282.7 37551 403.78

Step:  AIC=296.1
US_retailPrice ~ minifigs + depth + agerange_min + weight + rating

               Df Sum of Sq   RSS    AIC
<none>                       6426 296.10
- rating        1       681  7108 300.24
- minifigs      1       727  7154 300.64
- depth         1       728  7155 300.64
- agerange_min  1      5353 11779 331.06
- weight        1     31792 38218 402.85
```

```
summary(backward_stage2)
```

```
Call:
lm(formula = US_retailPrice ~ minifigs + depth + agerange_min +
    weight + rating, data = df_stage2)

Residuals:
    Min      1Q  Median      3Q     Max
-26.482  -5.907  -0.974   5.150  36.940

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.3865     7.4869  -0.052   0.9590
minifigs       2.7778     1.1134   2.495   0.0156 *
depth         -2.8752     1.1516  -2.497   0.0156 *
agerange_min   2.9673     0.4384   6.768 9.04e-09 ***
weight        58.8252     3.5662  16.495  < 2e-16 ***
rating        -1.8362     0.7603  -2.415   0.0191 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 10.81 on 55 degrees of freedom
Multiple R-squared:  0.9695,    Adjusted R-squared:  0.9667
F-statistic: 349.7 on 5 and 55 DF,  p-value: < 2.2e-16
```

```
formula(backward_stage2)
```

```
US_retailPrice ~ minifigs + depth + agerange_min + weight + rating
```

```
# Extract predictors retained by each procedure
best_stage1_terms <- attr(terms(model4), "term.labels")
backward_stage1_terms <- attr(
  terms(backward_stage1), "term.labels"
)

best_stage2_terms <- attr(terms(model5), "term.labels")
backward_stage2_terms <- attr(
  terms(backward_stage2), "term.labels"
)

# Show which predictors agree and which differ
list(
  Stage1_Best_Subsets = best_stage1_terms,
  Stage1_Backward = backward_stage1_terms,
  Stage1_Same_Model = setequal(
    best_stage1_terms, backward_stage1_terms
  ),
  Stage2_Best_Subsets = best_stage2_terms,
  Stage2_Backward = backward_stage2_terms,
  Stage2_Same_Model = setequal(
    best_stage2_terms, backward_stage2_terms
  )
)
```

```
$Stage1_Best_Subsets
[1] "pieces"   "minifigs" "height"   "depth"

$Stage1_Backward
[1] "pieces"   "minifigs" "height"   "depth"

$Stage1_Same_Model
[1] TRUE
```

```
$Stage2_Best_Subsets
[1] "minifigs"     "depth"        "agerange_min" "weight"       "rating"

$Stage2_Backward
[1] "minifigs"     "depth"        "agerange_min" "weight"       "rating"

$Stage2_Same_Model
[1] TRUE
```

```r
# Compare fit within each stage, using the same observations
data.frame(
  Stage = c("Stage 1", "Stage 1", "Stage 2", "Stage 2"),
  Method = c(
    "Best subsets", "Backward selection",
    "Best subsets", "Backward selection"
  ),
  AIC = c(
    AIC(model4), AIC(backward_stage1),
    AIC(model5), AIC(backward_stage2)
  ),
  Adjusted_R2 = c(
    summary(model4)$adj.r.squared,
    summary(backward_stage1)$adj.r.squared,
    summary(model5)$adj.r.squared,
    summary(backward_stage2)$adj.r.squared
  ),
  N = c(
    nobs(model4), nobs(backward_stage1),
    nobs(model5), nobs(backward_stage2)
  )
)
```

```
   Stage             Method       AIC Adjusted_R2   N
1 Stage 1       Best subsets 2532.9057   0.9568381 289
2 Stage 1 Backward selection 2532.9057   0.9568381 289
3 Stage 2       Best subsets  471.2063   0.9667275  61
4 Stage 2 Backward selection  471.2063   0.9667275  61
```

Backward Selection and Best-Subsets Model Comparison

The results show that backward selection produced exactly the same model as best subsets for both Stage 1 and Stage 2. For Stage 1, both procedures produced an AIC of 2532.91 and an adjusted R^2 of 0.9568 across 289 observations. Similarly, for Stage 2, both methods produced an AIC of 471.21 and an adjusted R^2 of 0.9667 across 61 observations. The high adjusted R^2 values indicate that both models explain a substantial proportion of the variation in LEGO set retail prices, with Stage 1 explaining approximately 95.7% and Stage 2 approximately 96.7% after accounting for the number of predictors. The identical results indicate that two different model-selection procedures converged on the same model for each stage, providing stronger evidence that the selected predictor combinations are appropriate choices among the models considered.

Stage 1 vs Stage 2 Comparison

The staged model-selection strategy was used to investigate the consequences of missing data. Because the models use different samples, their adjusted R-squared, Cp, and BIC values cannot be directly compared to decide which stage is better. These variables appear to have strong explanatory value, but requiring them reduces the usable sample from 1,079 sets to only 61 complete cases. By separating the analysis into two stages, we can evaluate how model conclusions change when richer predictors are included at the cost of dramatically reduced sample coverage.

Stage 1 uses predictors with relatively high coverage such as, pieces, minifigs, height, width, depth, and agerange_min, retaining 289 observations. This larger sample is more representative of the overall LEGO catalog and provides a stable foundation for model selection. Stage 2 requires weight, rating, and reviewCount, which reduces the dataset to 61 complete cases. This smaller subset includes larger, heavier sets with more complete data, introducing potential selection bias.

Predictor Strength Predictor strength differs between stages. In Stage 1, pieces is the strongest predictor, followed by depth, minifigs, height, and agerange_min. These variables reflect physical size and complexity, which are widely available and strongly associated with price.

In Stage 2, weight becomes the strongest predictor, followed by agerange_min, minifigs, depth, and rating. Weight enters the model first and remains included in every larger model, indicating that it captures information about set size and complexity that is not fully represented by piece count or dimensions. Rating and reviewCount enter only in larger models, suggesting weaker adjusted associations.

Model Selection Stage 1 selects a four-predictor model (pieces, minifigs, height, depth) as the best model. It has the highest adjusted R^2, lowest Cp, and lowest BIC, indicating strong

explanatory power with minimal complexity. Although five- and six-predictor models have similar adjusted R^2 values, their Cp and BIC values are higher, indicating more complexity.

Stage 2 selects a five-predictor model (weight, agerange_min, minifigs, depth, rating). Although the six-predictor model has the highest adjusted R^2, the difference is extremely small, and the five-predictor model has both the lowest Cp and lowest BIC, making it the preferred model.

Consequences of Missing Data The comparison highlights how missing values shapes model conclusions. Weight is a dominant predictor in Stage 2 but is missing for 893 sets in the full dataset. Ratings and review counts may reflect customer feedback collected after release. Stage 1 avoids these issues by focusing on predictors available for most sets, while Stage 2 explores the richer but selective subset.

Model Interpretation Stage 1 provides a broad model based on widely available physical characteristics. Stage 2 provides a richer model that incorporates weight and rating but is based on a much smaller and potentially selective sample. The staged approach allows us to compare these models directly and understand how predictor availability and sample coverage influence model selection. Stage 1 supports general pricing patterns across the dataset, while Stage 2 describes associations within a narrow subset of sets.

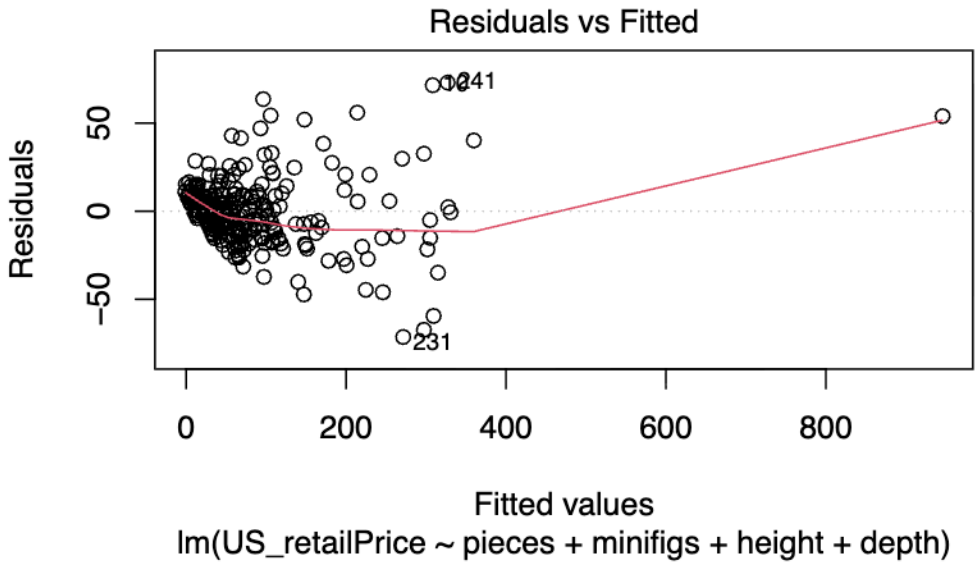
7. Model Diagnostics

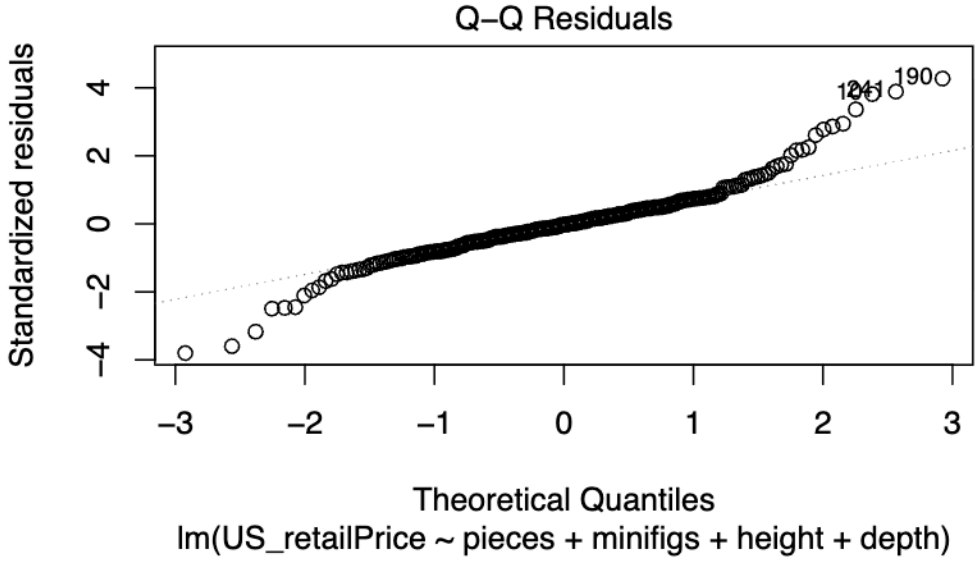
The diagnostic plots, residuals vs fitted, scale-location, q-q plot, cooks distance, will show whether the best subset model is actually valid.

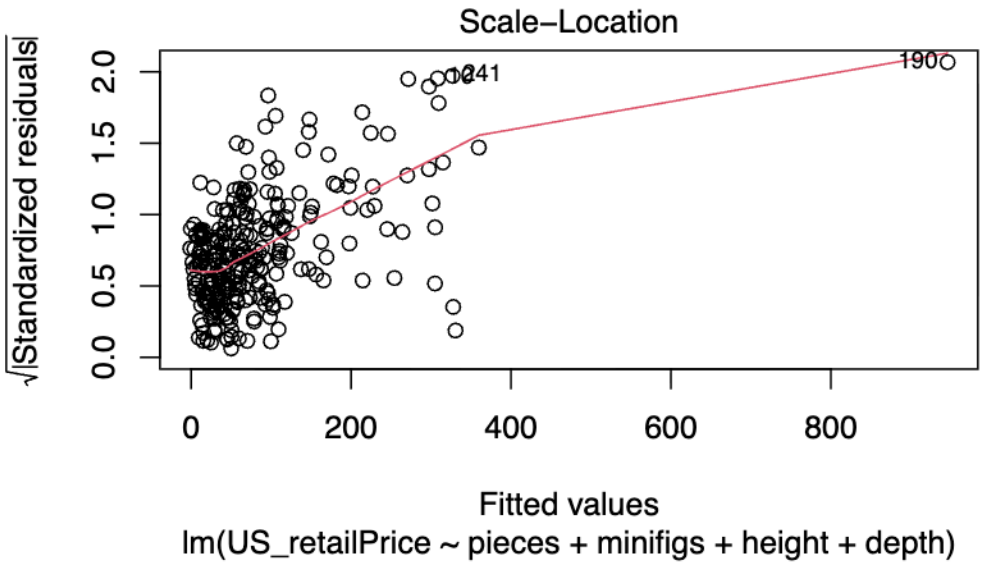
Stage 1

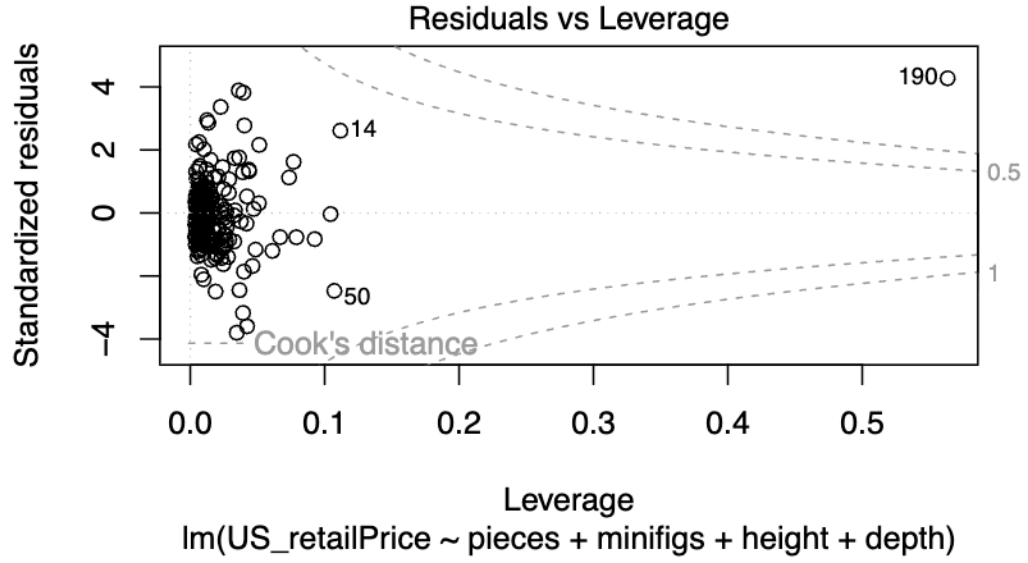
```
# fit stage 1 model 4
model4 <- lm(US_retailPrice ~ pieces + minifigs + height + depth,
             data = df_stage1)

# plots
par(mfrow = c(1,1))
plot(model4)
```


Residuals vs Fitted
Residuals
Fitted values
lm(US_retailPrice ~ pieces + minifigs + height + depth)


Q–Q Residuals
Standardized residuals
Theoretical Quantiles
lm(US_retailPrice ~ pieces + minifigs + height + depth)





In Stage 1, the residuals vs fitted plot shows a cluster of residuals around zero at lower fitted values, while the red LOESS line bends upward instead of remaining flat. This curvature is largely driven by a single influential observation with a fitted value near 900, which pulls the smoothing line upward and creates the appearance of a pattern. As a result, the residuals are

not entirely pattern-free.

The normal Q–Q plot indicates that the residuals follow the reference line closely through the center of the distribution, but the tails deviate, suggesting imperfect normality.

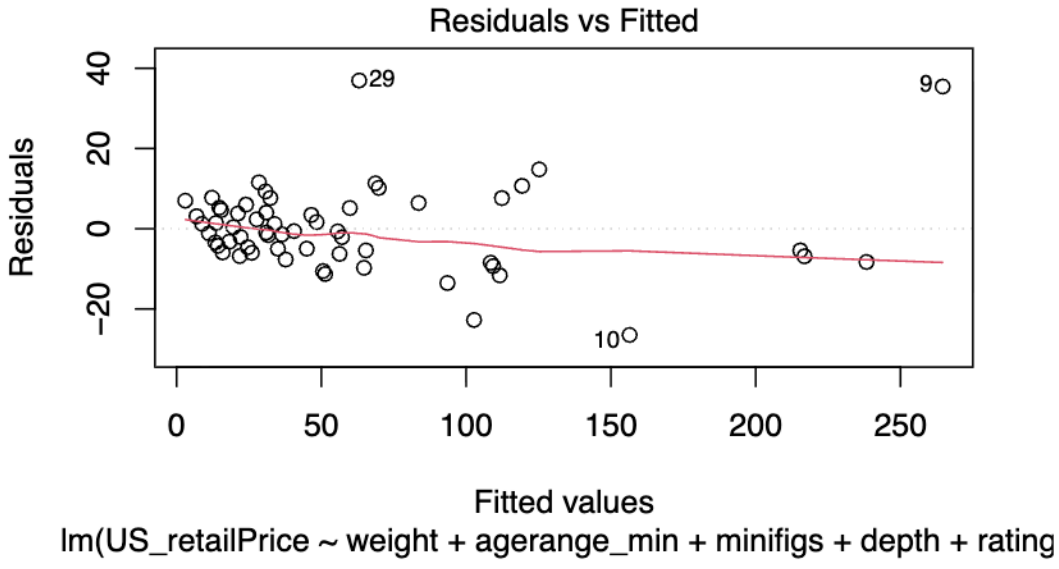
The scale–location plot, which displays the square root of the standardized residuals against fitted values, shows a rise in the red smoothing line on the right side of the plot. This suggests that variability may increase as fitted price increases, indicating possible nonconstant variance. However, the same high-leverage point appears to distort the trend and may exaggerate the apparent increase in spread.

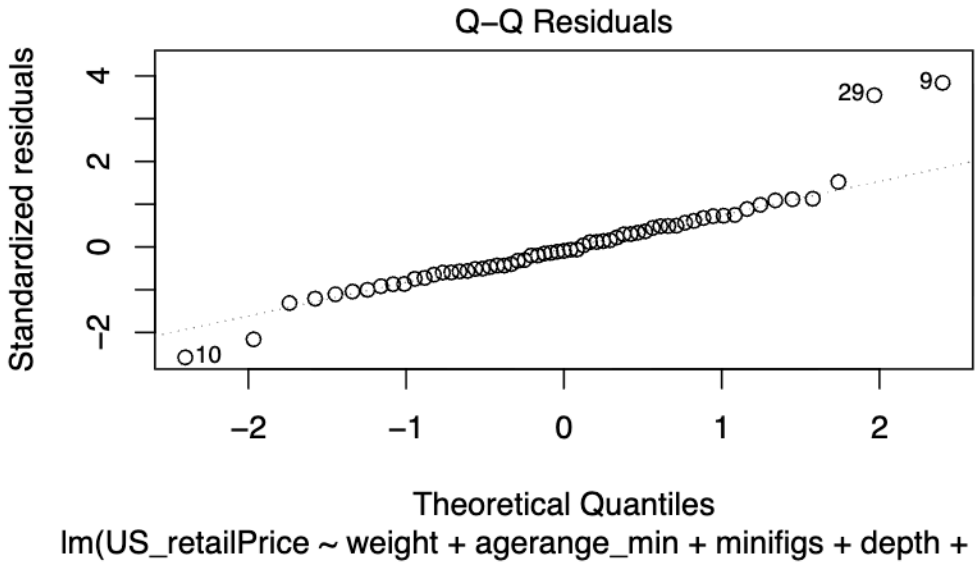
Finally, the residuals vs leverage plot confirms the presence of an observation with extremely high leverage. This point corresponds to LEGO set ID 51002 (Death Star), with a retail price of $899.99, affecting model inference.

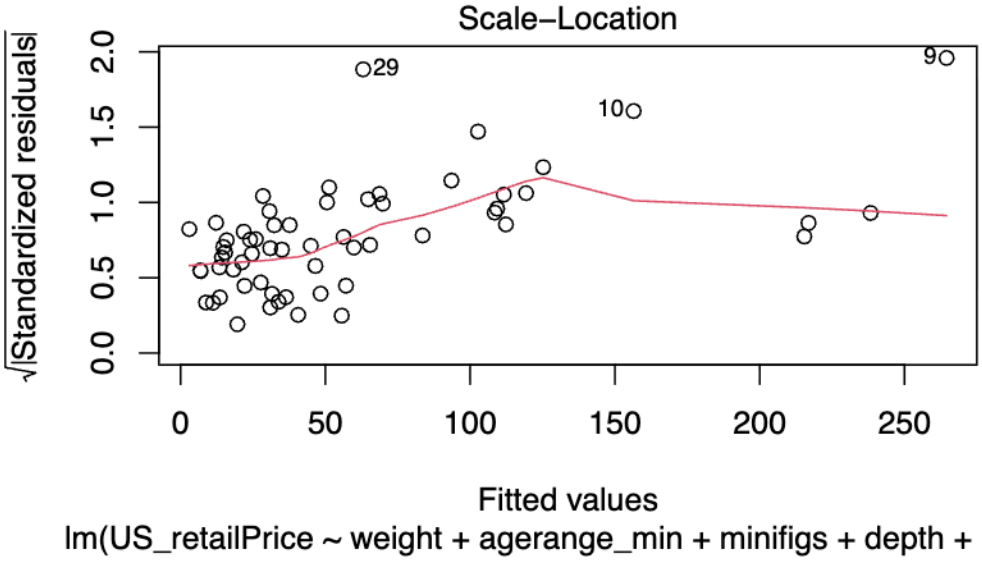
Stage 2

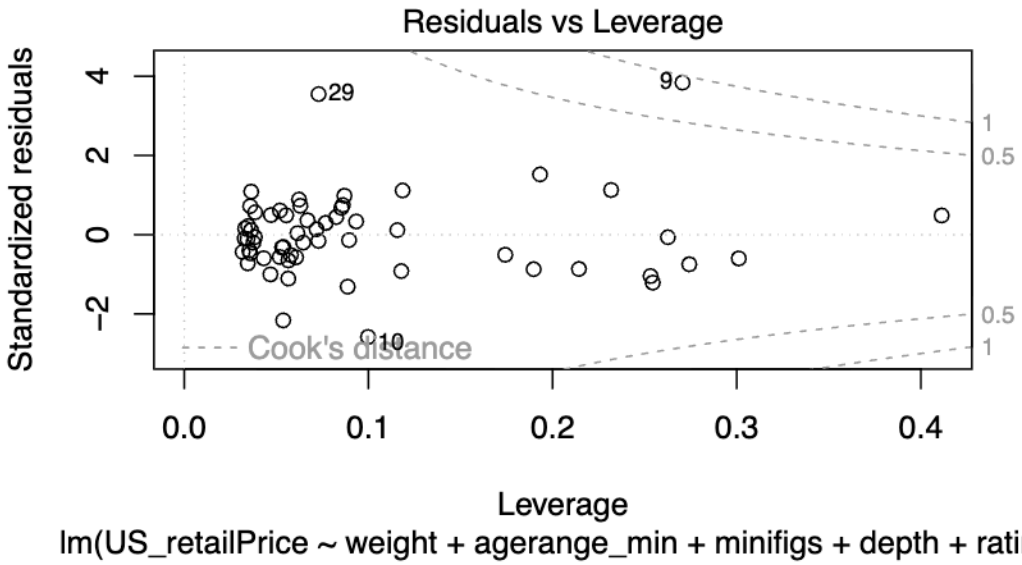
```
#fit stage 2 model 5
model5 <- lm(US_retailPrice ~ weight + agerange_min + minifigs + depth + rating,
             data = df_stage2)

#plots
par(mfrow = c(1,1))
plot(model5)
```




Q–Q Residuals
Standardized residuals
4
2
0
−2
29
9
10
−2
−1
0
1
2
Theoretical Quantiles
lm(US_retailPrice ~ weight + agerange_min + minifigs + depth + rating


Scale–Location
√|Standardized residuals|
2.0
1.5
1.0
0.5
0.0
29
10
9
0
50
100
150
200
250
Fitted values
lm(US_retailPrice ~ weight + agerange_min + minifigs + depth + rating



The Stage 2 diagnostic plots show some improvement in residual behavior, although several potential issues remain. In the residuals-versus-fitted plot, most observations are concentrated at lower fitted prices, while a few observations have relatively large residuals. The smooth curve shows some curvature, suggesting that the linear relationship may not fully capture every pattern in the data. The normal Q–Q plot follows the reference line reasonably well through the middle, but deviations in the tails suggest that the residuals may not be perfectly normally distributed. Similarly, the scale–location plot shows a slight upward trend before flattening, indicating that constant residual variance should not be assumed without further investigation.

The residuals versus leverage plot identifies observations 9, set ID 51221 Kakamora Barge listed at $99.99, and 29, set ID 51331 Italian Riviera listed at $299.99, as potentially unusual, with observation 9 showing particularly high leverage and a large standardized residual. These observations should be investigated using Cook's distance and, if appropriate, a sensitivity analysis to determine whether they substantially affect the model estimates. Overall, the Stage 2 diagnostics suggest that the model captures much of the relationship between the selected predictors and retail price, but possible nonlinearity, nonconstant variance, and influential observations remain concerns.

Diagnostic Comparison The Stage 1 and Stage 2 models both explain variations in LEGO retail price, but their diagnostic plots show that some assumptions may require further consideration, The Stage 1 Model uses more observations and provides a broader coverage of the dataset, although its plots how possible changes in residual variance and in unusual observation of high fitted values. The Stage 2 model has a more favorable residual pattern in

certain plots, but it still contains residuals with potential high leverage. The diagnostic plots do not establish that either model satisfies all regression assumptions, and with heteroskedasticity showing to be an issue in the residual plots, we will continue with a log transformation for Stage 1 and 2.

log transformations

```
df_stage1$log_price <- log(df_stage1$US_retailPrice)
df_stage2$log_price <- log(df_stage2$US_retailPrice)

# Stage 1: log-price model using the same predictors as model4
model4_log <- lm(
  log_price ~ pieces + minifigs + height + depth,
  data = df_stage1
)

# Stage 2: log-price model using the same predictors as model5
model5_log <- lm(
  log_price ~ weight + agerange_min + minifigs +
    depth + rating,
  data = df_stage2
)

summary(model4_log)
```

```
Call:
lm(formula = log_price ~ pieces + minifigs + height + depth,
    data = df_stage1)

Residuals:
     Min       1Q   Median       3Q      Max
-1.52052 -0.25026  0.02075  0.24439  0.85999

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.250e+00  9.863e-02  22.816  < 2e-16 ***
pieces       2.813e-04  5.475e-05   5.138 5.17e-07 ***
minifigs     1.966e-02  9.495e-03   2.071   0.0393 *
height       5.656e-02  3.182e-03  17.775  < 2e-16 ***
```

```
depth       -2.734e-03  1.409e-02  -0.194   0.8463
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3647 on 284 degrees of freedom
Multiple R-squared:  0.8283,    Adjusted R-squared:  0.8258
F-statistic: 342.4 on 4 and 284 DF,  p-value: < 2.2e-16
```

```r
summary(model5_log)
```

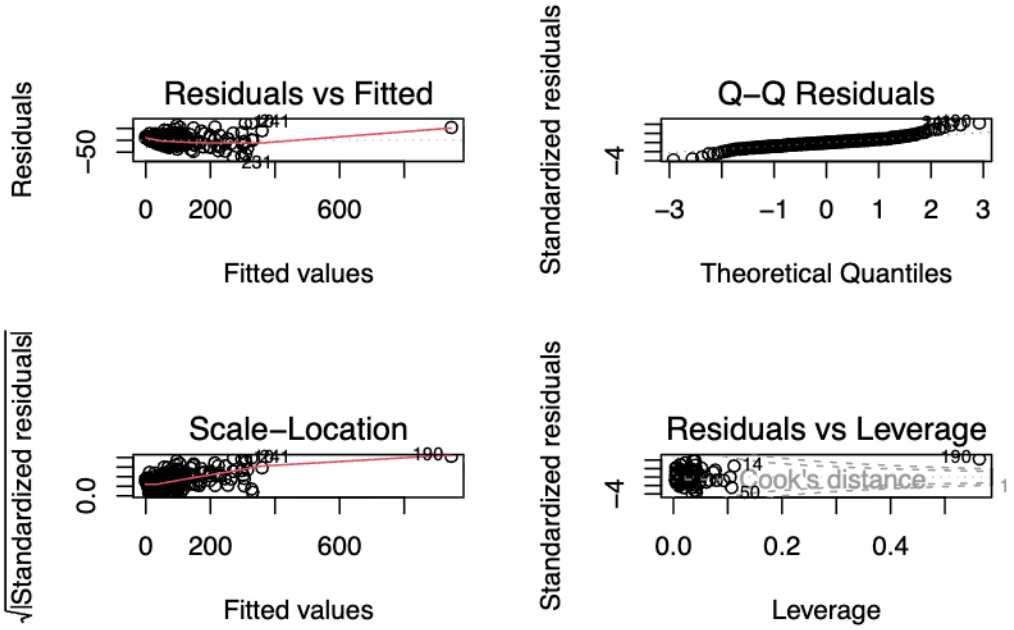
```
Call:
lm(formula = log_price ~ weight + agerange_min + minifigs + depth +
    rating, data = df_stage2)

Residuals:
     Min       1Q   Median       3Q      Max
-0.82553 -0.23675  0.08945  0.27985  0.97083

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.39163    0.29124   8.212 3.94e-11 ***
weight        0.59588    0.13872   4.295 7.15e-05 ***
agerange_min  0.04294    0.01705   2.518   0.0148 *
minifigs      0.07022    0.04331   1.621   0.1106
depth         0.04066    0.04480   0.908   0.3680
rating       -0.05149    0.02958  -1.741   0.0873 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

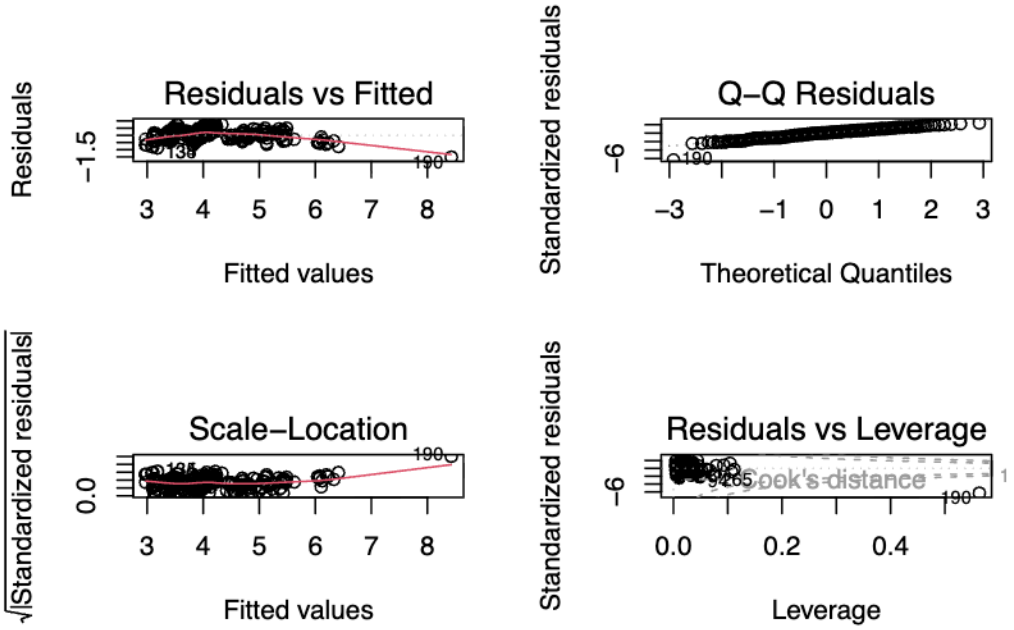
Residual standard error: 0.4205 on 55 degrees of freedom
Multiple R-squared:  0.7968,    Adjusted R-squared:  0.7784
F-statistic: 43.14 on 5 and 55 DF,  p-value: < 2.2e-16
```

```r
# Stage 1: compare the original and log-price models
par(mfrow = c(2, 2))
plot(model4)
```



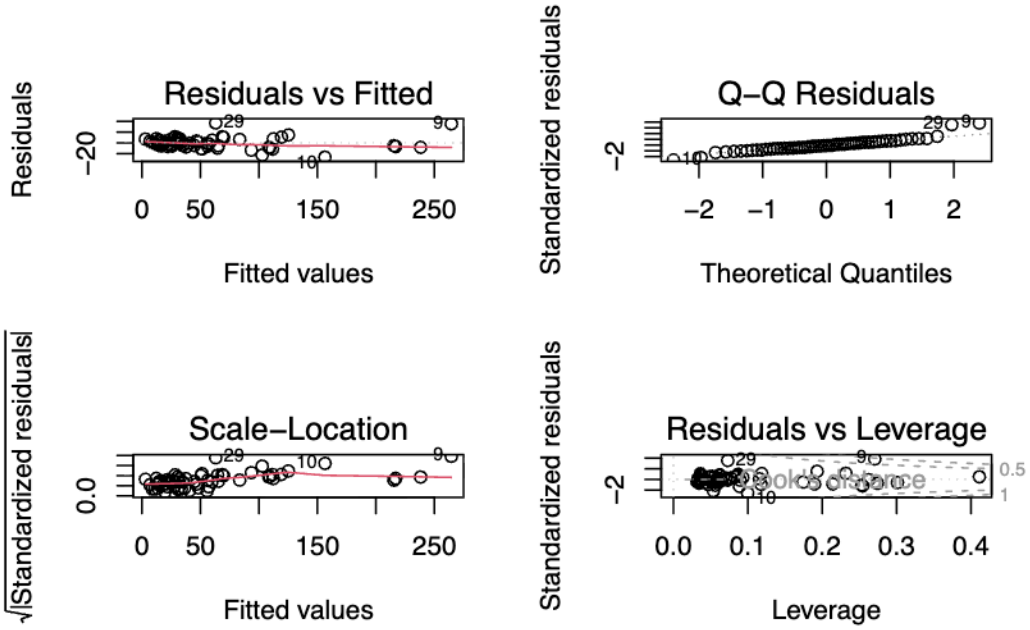

```
par(mfrow = c(1, 1))

par(mfrow = c(2, 2))
plot(model4_log)
```



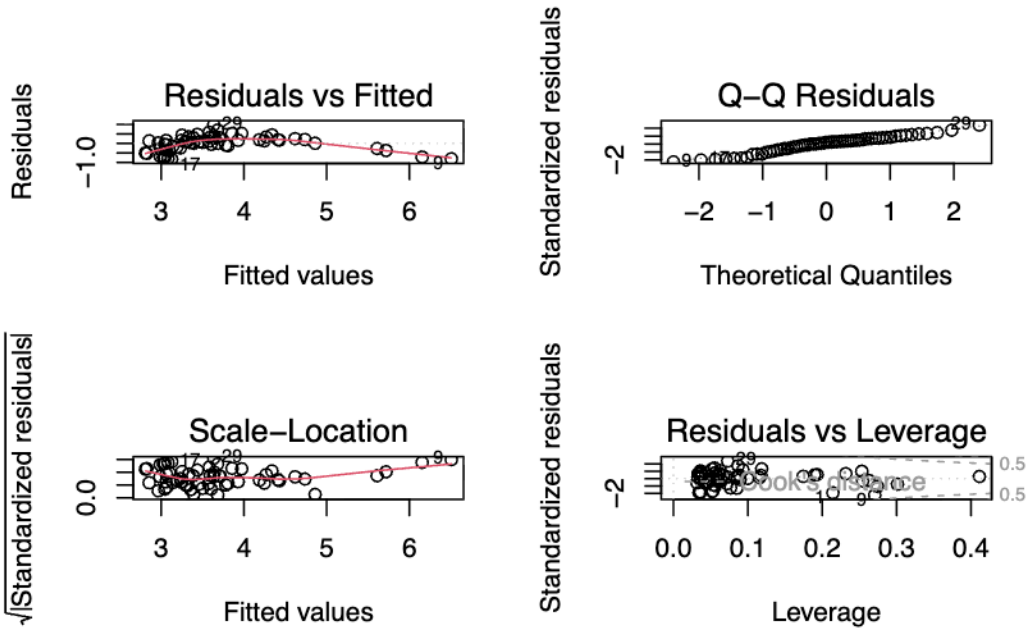

```
par(mfrow = c(1, 1))

# Stage 2: compare the original and log-price models
par(mfrow = c(2, 2))
plot(model5)
```




```
par(mfrow = c(1, 1))

par(mfrow = c(2, 2))
plot(model5_log)
```




```
par(mfrow = c(1, 1))
```

Analysis of Log-Transformed Regression Models

The diagnostic plots and regression results suggest that applying a logarithmic transformation to LEGO set retail price improves certain aspects of the regression models, particularly their ability to explain price variation and manage differences in scale. However, the transformation does not completely resolve issues with the linear regression assumptions, and its effectiveness differs between Stage 1 and Stage 2. The results also demonstrate that the characteristics most strongly associated with LEGO set prices depend on which observations and explanatory variables are included in each model.

Stage 1: Log-Transformed Model

The Stage 1 model uses 289 complete observations and predicts log-transformed retail price using pieces, minifigs, height, and depth. The model explains approximately 82.8% of the variation in log retail price, with an adjusted R^2 of 0.8258. The overall F-test is highly significant ($p<0.001$), indicating that the explanatory variables jointly provide substantial information about LEGO set prices.

The coefficient estimates suggest that piece count and height are the strongest statistically significant predictors in this model. Holding the other variables constant, piece count has

a positive and significant relationship with log price ($p<0.001$), as does height ($p<0.001$). Minifigures also have a positive, statistically significant association with price, although the evidence is weaker ($p=0.0393$). In contrast, depth is not statistically significant ($p=0.8463$), suggesting that it provides little additional explanatory information after accounting for the other variables. This may partly reflect overlap between depth and other measures of set size.

The diagnostic plots provide further insight into the model's performance:

Residuals vs. Fitted: The residuals are more concentrated around zero across much of the fitted-value range than might be expected with a poorly scaled response. However, the red smoothing line displays curvature, rising and then declining as fitted values increase. This suggests that some systematic structure remains in the residuals and that the relationships between the predictors and log price may not be fully linear. The model may therefore be missing nonlinear effects or interactions between set characteristics.

Normal Q-Q: The residuals follow a reasonably consistent progression through the center of the distribution but depart from the expected normal pattern in the tails. This indicates that the log transformation has not eliminated all departures from normality. Some LEGO sets may be substantially more or less expensive than the model predicts, even after accounting for their physical characteristics. These extreme residuals could reflect special editions, licensed themes, exclusive sets, or other pricing factors not included in the model.

Scale-Location: The red smoothing line decreases initially and then rises slightly at higher fitted values. This indicates that the spread of residuals is not perfectly constant. Nevertheless, the pattern is relatively moderate compared with a pronounced increase in residual spread, suggesting that the transformation may have helped stabilize variability across the price range. Some heteroscedasticity may remain, especially among sets with higher predicted prices.

Residuals vs. Leverage: Most observations have relatively low leverage, but several points stand apart from the main cluster. These observations have unusual combinations of explanatory-variable values and may influence the fitted regression. Although the plot does not establish that every such point is influential, it indicates that the model's estimates should be checked for sensitivity to individual observations.

Overall, Stage 1 demonstrates that the log-transformed model captures a strong relationship between LEGO set characteristics and retail price. The high adjusted R^2, significant overall F-test, and significant coefficients for pieces, height, and minifigs support the model's explanatory value. However, the remaining curvature, nonconstant residual spread, and tail departures from normality suggest that the transformation improves the model without fully satisfying all regression assumptions.

Stage 2: Log-Transformed Model

The Stage 2 model uses a different subset of the data, with 61 observations, and predicts log retail price using weight, minimum age range, minifigs, depth, and rating. Its adjusted R^2 is

0.7784, meaning that the model explains approximately 77.8% of the variation in log price after accounting for the number of predictors. The overall F-test is highly significant ($p<0.001$), demonstrating that the predictors collectively have a statistically significant relationship with retail price.

However, the individual coefficient results are more mixed than those in Stage 1. Weight has a positive and statistically significant association with log price ($p<0.001$), as does minimum age range ($p=0.0148$). This suggests that heavier LEGO sets tend to have higher prices and that sets designed for older age groups may also command higher prices, after controlling for the other variables. Minifigures ($p=0.1106$) and depth ($p=0.3680$) are not statistically significant at the 5% level. Rating is marginally significant at the 10% level ($p=0.0873$), with a negative coefficient, but the evidence is insufficient to conclude that ratings have a reliable independent association with price.

The diagnostic plots reveal several important limitations:

Residuals vs. Fitted: The residuals are concentrated around zero for many fitted values, but the plot includes several observations with very large fitted values relative to the majority. The red smoothing line also shows some curvature. This suggests that the model may not capture the relationship between all predictor combinations and log price equally well. The concentration of most observations at lower fitted values, alongside a small number of high-value predictions, also indicates that the model is being fitted to a dataset with a limited and uneven distribution of set characteristics.

Normal Q-Q: The residuals depart from the reference pattern, particularly in the tails. This suggests that the residual distribution remains non-normal despite transforming price. A few observations appear particularly unusual, including observations with strongly negative standardized residuals. Such points may represent LEGO sets whose prices differ substantially from the values predicted by their recorded characteristics. The small sample size also makes the model more vulnerable to individual observations that do not follow the general pattern.

Scale-Location: The red smoothing line increases across much of the fitted-value range, with a more pronounced rise at the upper end. This indicates that residual variability tends to increase as fitted values increase. Consequently, the log transformation has not fully resolved heteroscedasticity in Stage 2. The small number of observations at higher fitted values may contribute to this pattern, since a few expensive or unusual sets can have a disproportionate effect on the apparent residual spread.

Residuals vs. Leverage: Several observations have noticeably greater leverage than the main cluster, including a point at the extreme right of the plot. These observations have unusual predictor combinations and may have substantial influence on the estimated coefficients. The plot therefore raises concerns about model stability, particularly because the dataset contains only 61 observations. The effect of these observations should be investigated using Cook's distance and sensitivity analyses rather than assuming that all high-leverage points are necessarily problematic.

Taken together, the Stage 2 results indicate that the model explains a substantial proportion of price variation, but its smaller sample and remaining diagnostic issues limit confidence in its individual coefficient estimates. The statistically significant overall model does not mean that every predictor is independently useful, nor does the relatively high adjusted R^2 establish that the model satisfies the assumptions required for conventional regression inference.

As such, while taking the log transformations of both Stage 1 and Stage 2 helped with controlling certain aspects, we retained the original-price models rather than the log-transformed models because the original models provided a more direct interpretation of the relationship between LEGO set characteristics and U.S. retail price, with coefficients expressed in dollars. Although the log transformation was explored to address potential heteroskedasticity, the diagnostic plots still showed some residual patterns, suggesting that the transformation did not fully resolve the model assumptions. Therefore, we retained the original models for interpretability, while acknowledging that some diagnostic concerns remain.

8. Final Model Selection

The staged model-selection process produced two candidate models: a high-coverage Stage 1 model based on 289 observations and a richer Stage 2 model based on 61 complete cases. Stage 1 identified a four-predictor model including pieces, minifigs, height, and depth as the best model according to adjusted R-squared, Cp, and BIC. Stage 2 identified a five-predictor model including weight, agerange_min, minifigs, depth, and rating as the best model, with the lowest Cp and BIC and an adjusted R-squared nearly identical to the larger six-predictor alternative.

Both models were retained for interpretation because they are fit on different samples, and their fit statistics cannot be compared as though they were calculated from the same data. The final choice therefore reflects the selection criteria within each stage and the intended focus on the richer predictor set available in Stage 2, rather than evidence that Stage 2 is objectively "better" than Stage 1.

Regression Equation Interpretation

```
#regression equations
summary(model5)
```

```
Call:
lm(formula = US_retailPrice ~ weight + agerange_min + minifigs +
    depth + rating, data = df_stage2)
```

```
Residuals:
    Min      1Q  Median      3Q     Max
-26.482  -5.907  -0.974   5.150  36.940

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.3865     7.4869  -0.052   0.9590
weight        58.8252     3.5662  16.495  < 2e-16 ***
agerange_min   2.9673     0.4384   6.768 9.04e-09 ***
minifigs       2.7778     1.1134   2.495   0.0156 *
depth         -2.8752     1.1516  -2.497   0.0156 *
rating        -1.8362     0.7603  -2.415   0.0191 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.81 on 55 degrees of freedom
Multiple R-squared:  0.9695,    Adjusted R-squared:  0.9667
F-statistic: 349.7 on 5 and 55 DF,  p-value: < 2.2e-16
```

```
summary(model4)
```

```
Call:
lm(formula = US_retailPrice ~ pieces + minifigs + height + depth,
    data = df_stage1)

Residuals:
    Min      1Q  Median      3Q     Max
-71.426  -9.972  -0.393   8.761  73.052

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -48.675643   5.174107  -9.408  < 2e-16 ***
pieces        0.060727   0.002872  21.144  < 2e-16 ***
minifigs      4.301790   0.498088   8.637 4.25e-16 ***
height        0.939902   0.166915   5.631 4.30e-08 ***
depth         5.887810   0.739335   7.964 4.05e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.13 on 284 degrees of freedom
Multiple R-squared:  0.9574,    Adjusted R-squared:  0.9568
```

```
F-statistic:  1597 on 4 and 284 DF,  p-value: < 2.2e-16
```

```
# load package and check for collinearity
library(car)
```

```
Loading required package: carData
```

```
Attaching package: 'car'
```

```
The following object is masked from 'package:dplyr':

    recode
```

```
vif(model5)
```

```
      weight agerange_min     minifigs        depth       rating
    5.608980     1.375048     2.448352     3.415140     1.177649
```

```
vif(model4)
```

```
  pieces minifigs   height    depth
5.416017 2.293975 2.084717 4.070719
```

Stage 1 Final Regression Equation:

$$\text{US } \widehat{\text{retail}} \text{ price} = -48.675 + 0.0607\,(\text{pieces}) + 4.3017\,(\text{minifigs}) + 0.09399\,(\text{height}) + 5.887\,(\text{depth})$$

The intercept of -48.675 represents the predicted price when all predictors are zero. Not meaningful substantively but is included for model completeness.

Pieces (+0.0607) holding all other predictors constant, each additional piece is associated with an estimated $0.06 increase in retail price.

Minifigs (+4.3017) holding all the other predictors constant, each additional minifigure in minimum age recommended increases retail price by $4.30

Height (+0.09399) for each one-unit increase in height, retail price increases by $0.09, holding other predictors constant.

Depth (+5.887) for each additional one-unit of package depth retail, price increases by $5.89, holding other predictors constant.

VIF values were all below 10, indicating no severe multicollinearity. Pieces (5.42) and depth (4.07) showed moderately elevated VIFs due to overlap with other physical size variables, but these levels do not compromise model stability.

Stage 2 Final Regression Equation:

$$\begin{aligned}\widehat{\text{US retail price}} = & -0.3865 \\ & +58.8252\,(\text{weight}) \\ & +2.9673\,(\text{agerange_min}) \\ & +2.7778\,(\text{minifigs}) \\ & -2.8752\,(\text{depth}) \\ & -1.8362\,(\text{rating})\end{aligned}$$

The intercept of -0.3865 represents the predicted price when all predictors are zero. Not meaningful substantively but is included for model completeness.

Weight (+58.83) holding all other predictors constant, each additional one-unit of weight is associated with an estimated $58.83 increase in retail price. Weight has the strongest statistical evidence of an association with price in this model, with a very small p-value. This aligns with intuition: heavier sets contain more material, larger builds, and higher production costs.

Minimum Age (+2.97) holding all the other predictors constant, each additional year in minimum age recommended increases retail price by $2.97.

Minifigs (+2.78) each additional minifigure is associated with an increase in retail price by $2.78, holding other predictors constant.

Depth (–2.88) for each additional one-unit of package depth retail, price decreases by $2.88, holding other predictors constant.

Rating (–1.84) for each one-unit increase in customer rating, retail price decreases by $1.84, holding other predictors constant.

VIF values were all below 10, indicating no severe multicollinearity. Weight had a moderately elevated VIF (5.61), reflecting its overlap with other physical size variables, but this does not compromise model stability.

To ensure no final issues with multicollinearity, we will investigate whether dropping the two variables, pieces and weight, with a VIF over 5 makes the model worse.

```
library(car)

# Original Stage 1 model
model4 <- lm(
  US_retailPrice ~ pieces + minifigs + height + depth,
```

```
  data = df_stage1
)

# Alternative model without pieces
model4_no_pieces <- lm(
  US_retailPrice ~ minifigs + height + depth,
  data = df_stage1
)

# Compare model fit
stage1_comparison <- data.frame(
  Model = c("With pieces", "Without pieces"),
  N = c(nobs(model4), nobs(model4_no_pieces)),
  R_squared = c(
    summary(model4)$r.squared,
    summary(model4_no_pieces)$r.squared
  ),
  Adjusted_R_squared = c(
    summary(model4)$adj.r.squared,
    summary(model4_no_pieces)$adj.r.squared
  ),
  AIC = c(AIC(model4), AIC(model4_no_pieces)),
  Residual_SE = c(sigma(model4), sigma(model4_no_pieces))
)

stage1_comparison
```

```
           Model   N R_squared Adjusted_R_squared      AIC Residual_SE
1    With pieces 289 0.9574376          0.9568381 2532.906    19.12921
2 Without pieces 289 0.8904352          0.8892819 2804.168    30.63769
```

```
# Check whether multicollinearity improves
vif(model4)
```

```
  pieces minifigs   height    depth
5.416017 2.293975 2.084717 4.070719
```

```
vif(model4_no_pieces)
```

```
minifigs   height    depth
1.969467 1.788633 2.494103
```

```r
# Examine the models in detail
summary(model4)
```

```
Call:
lm(formula = US_retailPrice ~ pieces + minifigs + height + depth,
    data = df_stage1)

Residuals:
    Min      1Q  Median      3Q     Max
-71.426  -9.972  -0.393   8.761  73.052

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -48.675643   5.174107  -9.408  < 2e-16 ***
pieces        0.060727   0.002872  21.144  < 2e-16 ***
minifigs      4.301790   0.498088   8.637 4.25e-16 ***
height        0.939902   0.166915   5.631 4.30e-08 ***
depth         5.887810   0.739335   7.964 4.05e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.13 on 284 degrees of freedom
Multiple R-squared:  0.9574,    Adjusted R-squared:  0.9568
F-statistic:  1597 on 4 and 284 DF,  p-value: < 2.2e-16
```

```r
summary(model4_no_pieces)
```

```
Call:
lm(formula = US_retailPrice ~ minifigs + height + depth, data = df_stage1)

Residuals:
     Min       1Q   Median       3Q      Max
-140.881  -18.384    0.344   17.347   97.081

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -127.2308     5.7677 -22.059   <2e-16 ***
minifigs       8.2629     0.7392  11.179   <2e-16 ***
height         2.2700     0.2476   9.167   <2e-16 ***
```

```
depth          15.6166     0.9269  16.849   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.64 on 285 degrees of freedom
Multiple R-squared:  0.8904,    Adjusted R-squared:  0.8893
F-statistic: 772.1 on 3 and 285 DF,  p-value: < 2.2e-16
```

```r
# Original Stage 2 model
model5 <- lm(
  US_retailPrice ~ weight + agerange_min +
    minifigs + depth + rating,
  data = df_stage2
)

# Alternative model without weight
model5_no_weight <- lm(
  US_retailPrice ~ agerange_min + minifigs +
    depth + rating,
  data = df_stage2
)

# Compare model fit
stage2_comparison <- data.frame(
  Model = c("With weight", "Without weight"),
  N = c(nobs(model5), nobs(model5_no_weight)),
  R_squared = c(
    summary(model5)$r.squared,
    summary(model5_no_weight)$r.squared
  ),
  Adjusted_R_squared = c(
    summary(model5)$adj.r.squared,
    summary(model5_no_weight)$adj.r.squared
  ),
  AIC = c(AIC(model5), AIC(model5_no_weight)),
  Residual_SE = c(sigma(model5), sigma(model5_no_weight))
)

stage2_comparison
```

```
           Model  N R_squared Adjusted_R_squared      AIC Residual_SE
1    With weight 61 0.9695002          0.9667275 471.2063    10.80949
```

```
2 Without weight 61 0.8186168          0.8056608 577.9628    26.12419
```

```r
# Check whether multicollinearity improves
vif(model5)
```

```
      weight agerange_min     minifigs        depth       rating
    5.608980     1.375048     2.448352     3.415140     1.177649
```

```r
vif(model5_no_weight)
```

```
agerange_min     minifigs        depth       rating
    1.365078     1.497791     1.493132     1.169487
```

```r
# Examine the models in detail
summary(model5)
```

```
Call:
lm(formula = US_retailPrice ~ weight + agerange_min + minifigs +
    depth + rating, data = df_stage2)

Residuals:
    Min      1Q  Median      3Q     Max
-26.482  -5.907  -0.974   5.150  36.940

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.3865     7.4869  -0.052   0.9590
weight        58.8252     3.5662  16.495  < 2e-16 ***
agerange_min   2.9673     0.4384   6.768 9.04e-09 ***
minifigs       2.7778     1.1134   2.495   0.0156 *
depth         -2.8752     1.1516  -2.497   0.0156 *
rating        -1.8362     0.7603  -2.415   0.0191 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.81 on 55 degrees of freedom
Multiple R-squared:  0.9695,    Adjusted R-squared:  0.9667
F-statistic: 349.7 on 5 and 55 DF,  p-value: < 2.2e-16
```

```r
summary(model5_no_weight)
```

```
Call:
lm(formula = US_retailPrice ~ agerange_min + minifigs + depth +
    rating, data = df_stage2)

Residuals:
    Min      1Q  Median      3Q     Max
-76.713 -11.502   1.576   9.212  72.586

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -94.3227    11.7465  -8.030 6.91e-11 ***
agerange_min   3.5830     1.0557   3.394  0.00127 **
minifigs      14.2209     2.1046   6.757 8.71e-09 ***
depth         11.3754     1.8403   6.181 7.70e-08 ***
rating        -0.7921     1.8312  -0.433  0.66700
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 26.12 on 56 degrees of freedom
Multiple R-squared:  0.8186,    Adjusted R-squared:  0.8057
F-statistic: 63.18 on 4 and 56 DF,  p-value: < 2.2e-16
```

```r
# Does removing pieces significantly worsen Stage 1 fit?
anova(model4_no_pieces, model4)
```

```
Analysis of Variance Table

Model 1: US_retailPrice ~ minifigs + height + depth
Model 2: US_retailPrice ~ pieces + minifigs + height + depth
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    285 267520
2    284 103923  1    163597 447.08 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Does removing weight significantly worsen Stage 2 fit?
anova(model5_no_weight, model5)
```

```
Analysis of Variance Table

Model 1: US_retailPrice ~ agerange_min + minifigs + depth + rating
Model 2: US_retailPrice ~ weight + agerange_min + minifigs + depth + rating
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)
1     56 38218
2     55  6426  1     31792 272.09 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Sensitivity Analysis for Multicollinearity

To investigate the potential consequences of multicollinearity, we conducted a sensitivity analysis by removing the predictor with a VIF greater than 5 from each selected model. In Stage 1, pieces had a VIF of approximately 5.42. Removing it reduced R^2 from 0.9574 to 0.8904 and adjusted R^2 from 0.9568 to 0.8893. AIC increased from 2,532.91 to 2,804.17, and a nested-model F-test indicated a statistically significant loss of explanatory information. These results suggest that pieces contributes substantial explanatory information beyond the other predictors in the Stage 1 model.

Similarly, in Stage 2, weight had a VIF of approximately 5.61. Removing it reduced R^2 from 0.9695 to 0.8186 and adjusted R^2 from 0.9667 to 0.8057. AIC increased from 471.21 to 577.96, while the nested-model F-test indicated a statistically significant loss of explanatory information. These results suggest that weight contributes substantial explanatory information beyond the other predictors in the Stage 2 model.

Overall, removing either high-VIF predictor substantially reduced model fit. Although pieces and weight exhibit some multicollinearity with other predictors, the sensitivity analysis indicates that both contribute important explanatory information to their respective models. We therefore retained both predictors rather than removing them solely to reduce their VIFs. This decision balances the concern about multicollinearity against the substantial loss of explanatory power that would result from excluding these variables.

9. Limitations

Several limitations should be considered when interpreting the results of this analysis. First, the dataset is restricted to LEGO sets released in 2025, limiting the findings to the characteristics and pricing patterns of sets within that release year. Consequently, the results may not generalize to LEGO sets released in other years, when product designs, pricing strategies, and market conditions may differ. Additionally, the analysis examines original U.S. retail prices rather than resale values or current market prices, meaning the findings do not account for changes in value after a set's initial release. In reality, LEGO sets frequently change in value

after being released thus our models may not reflect the lived reality of what factors are most strongly associated with general price.

A second limitation concerns missing data and the use of two analytical stages. Stage 2 uses only observations with complete data across all variables included in the analysis, whereas Stage 1 uses the broader dataset, allowing observations to be retained even when some variables have missing values. This approach allows the analysis to compare results across different data samples and examine whether the conclusions remain consistent. However, Stage 2's complete-case requirement reduces the sample size and may introduce bias if observations with missing values differ systematically from those with complete data. Although Stage 1 incorporates more of the available dataset, the inclusion of additional observations does not automatically eliminate the limitations associated with missing data. The available information may be insufficient to determine whether missingness is random or related to particular set characteristics. Furthermore, differences between the two stages may reflect changes in sample composition rather than genuine differences in the relationships between predictors and retail price. Therefore, comparisons between the stages must account for the different data used in each analysis.

Another important limitation is the influence of an individual observation on the regression results. In both Stage 1 and Stage 2, one influential observation affected the residual structure, suggesting that the models may be sensitive to unusual observations. Influential points can affect estimated regression coefficients, model fit, and the apparent adequacy of regression assumptions. Although such observations may represent legitimate LEGO sets rather than data errors, their influence raises questions about the stability of the selected models. As a result, the findings should be interpreted cautiously, and the influential observation should be examined to determine whether the conclusions change substantially when its influence is assessed.

Finally, the analysis relies on observational data and linear regression methods. Although model selection techniques help identify combinations of variables that explain variation in retail price, they do not establish causal relationships. Some relevant characteristics may be unavailable in the dataset, while relationships between price and the included predictors may be nonlinear or influenced by interactions that the selected models do not capture. Consequently, the final model should be interpreted as an estimate of the associations between observed LEGO set characteristics and original U.S. retail price, rather than a definitive explanation of how LEGO determines its prices.

10. Conclusion

This project used multiple linear regression, backwards selection, and best-subsets model selection to examine how LEGO set characteristics are associated with US retail price. Because the original dataset contained substantial missing data, we used a two-stage approach to evaluate models with different levels of predictor availability. Stage 1 used 289 observations

and selected a four-predictor model containing piece count, minifigure count, height, and depth. Stage 2 incorporated additional information—particularly weight and customer rating—but was limited to 61 complete observations.

The final model selected is the Stage 2 five-predictor model:

$$\begin{aligned}\widehat{\text{US retail price}} = &-0.3865\\ &+58.8252\,(\text{weight})\\ &+2.9673\,(\text{agerange_min})\\ &+2.7778\,(\text{minifigs})\\ &-2.8752\,(\text{depth})\\ &-1.8362\,(\text{rating})\end{aligned}$$

This model balances explanatory strength, diagnostic stability, offering an understanding on how physical attributes and customer sentiment relate to LEGO pricing. While limitations related to missingness, sample size, and potential selection bias remain, the staged approach provides a strong and interpretable analysis of the factors influencing retail price.

These results should be interpreted with several limitations in mind. Requiring complete data reduced the analysis from the original 1,079 sets to only 61 observations in Stage 2, which may limit how well the final model represents the full LEGO catalog. The diagnostic plots also identified unusual residuals and a potentially high-leverage observation. In addition, the analysis identifies associations rather than causal effects, and the relationships observed in the selected sample may not generalize to all LEGO sets.

Overall, the analysis demonstrates that physical characteristics and other set attributes can collectively explain substantial variation in LEGO retail price within the analyzed sample. The staged approach made it possible to compare a broader model with a richer but smaller model, showing the trade-off between sample coverage and the availability of additional predictors. Further investigation of missing data, influential observations, and model assumptions would help determine how reliably these relationships extend to the wider LEGO catalog.